LEVERAGING PREDICTABILITY OF DEEP LEARNING TECHNIQUES ON FINANCIAL TIME SERIES FORECASTING

A Project Report submitted in partial fulfillment of the requirements for the award of

The degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

Submitted by
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DECLARATION

I hereby declare that the project report entitled "Leveraging Predictability of Deep Learning Techniques on Financial Time Series Forecasting" is an original work done in the Department of Computer Science and Engineering, GITAM School of Technology, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the project report entitled "Leveraging Predictability of Deep Learning Techniques on Financial Time Series Forecasting " is a bonafide record of work carried out by Sri Vaishnavi Vasa (HU21CSEN0100673), Leena Deekollu (HU21CSEN0100741), Y. Bindu Sri (HU21CSEN0101184) students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering.

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Sincerely,

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CHAPTER 1: ABSTRACT

Financial time series forecasting remains a challenging task due to the inherent volatility and unpredictability of markets, particularly in emerging economies such as India. Traditional statistical models and machine learning approaches struggle to capture the complex, non-linear dependencies in financial data. In this study, we propose a model integrating Bidirectional Long Short-Term Memory (Bi-LSTM) and Dense Networks, to enhance predictive accuracy. The Bi-LSTM component effectively captures sequential dependencies and long-term trends, while the Dense branch extracts meaningful non-temporal relationships, leading to a richer representation of market patterns.

Extensive experimentation with baseline, hybrid deep learning models, and probabilistic approaches demonstrated the challenges posed by fluctuating stock market dynamics. To address model interpretability, we incorporate Explainable AI (XAI) techniques, ensuring transparency in feature significance and model decisions. Our results highlight the advantages of combining sequential and non-sequential learning paradigms for financial forecasting, providing deeper insights into stock market trends. This research contributes to the growing field of AI-driven financial modeling by identifying an effective approach for handling high-variance stock market data.

CHAPTER 2: INTRODUCTION

2.1 BACKGROUND AND MOTIVATION

Time series forecasting is the process of predicting future values based on previously observed data points collected over time. Time series data is a sequence of observations recorded at regular intervals, such as daily stock prices, monthly sales figures, or yearly economic indicators. Unlike standard datasets, time series data exhibits temporal dependencies, meaning that past values influence future outcomes. Various forecasting methods, ranging from traditional statistical models like ARIMA to modern deep learning techniques, are employed to analyze and predict trends in time-dependent data.

Financial time series forecasting focuses specifically on predicting stock prices, currency exchange rates, and other financial indicators. This task is inherently challenging due to the high volatility, non-linearity, and external influences such as market sentiment, economic policies, and geopolitical events. While conventional models attempt to capture historical trends, deep learning (DL) models have shown promise in identifying complex, non-linear patterns in financial data, improving prediction accuracy.

This project explores deep learning-based approaches for financial time series forecasting, particularly for fluctuating markets like the Indian Stock Exchange. Through experimentation with various baseline and hybrid DL models, we aim to enhance forecasting performance by integrating Bidirectional LSTM (Bi-LSTM) for capturing sequential dependencies and a Dense Network for learning non-sequential patterns. Additionally, we incorporate Explainable AI (XAI) to improve model interpretability, ensuring that predictions are transparent and meaningful for financial analysts.

2.2 PROBLEM DEFINITION

Time series data is a sequence of observations recorded at regular time intervals, often used to analyze patterns, trends, and seasonality over time. Mathematically, it is represented as $Y_t=f(t)+\epsilon_t$, where Y_t is the observed value at time t, f(t) is the underlying trend, and ϵt\epsilon_tϵt is the random error.

Time series forecasting aims to predict future values based on historical data. A common approach is the autoregressive (AR) model, expressed as $Y_t=\phi_1 Y_{t-1}+\phi_2 Y_{t-2}+\cdots+\phi_p Y_{t-p}+\epsilon_t$, where Y_t depends on past observations $Y_{t-1}, Y_{t-2}, \ldots, Y_{t-p}$, ϕ_i are the model parameters, and ϵ_t represents white noise.

2.3 OBJECTIVES

The primary objective of this project is to develop an improved deep learning-based financial time series forecasting model that effectively captures complex market patterns while ensuring transparency and interpretability. Given the high volatility and unpredictability of financial markets, particularly in emerging economies like India, traditional forecasting models often fail to generalize well. This study aims to bridge the gap between predictive accuracy and explainability by integrating advanced deep learning techniques with Explainable AI (XAI).

The specific objectives of this research are as follows:

- To enhance the accuracy of financial time series forecasting by leveraging a hybrid deep learning approach that combines Bidirectional Long Short-Term Memory (Bi-LSTM) for sequential dependencies and Dense Networks for learning non-sequential relationships.
- To improve model robustness in fluctuating market conditions by experimenting with various baseline and hybrid models, identifying the most effective architecture for forecasting stock prices.
- To incorporate Explainable AI (XAI) techniques to evaluate the significance of different features and provide transparent insights into the model's decision-making process, making predictions more interpretable for financial analysts and investors.

- To evaluate and compare the proposed model against existing statistical, machine learning, and deep learning approaches to assess its effectiveness in capturing financial market trends.

By achieving these objectives, this project contributes to the development of a more accurate, robust, and interpretable AI-driven financial forecasting model, aiding market participants in making well-informed decisions.

2.4 SCOPE

This project focuses on forecasting stock prices within the Indian financial market, specifically using historical data from the Bombay Stock Exchange (BSE) SENSEX NEXT50 from 2014 to 2024. This dataset encompasses major market events and fluctuations, allowing the model to learn from diverse financial conditions. The study explores the effectiveness of deep learning techniques, particularly a hybrid Bi-LSTM + Dense Network model, in capturing both sequential and non-sequential dependencies in financial time series data. Additionally, Explainable AI (XAI) is incorporated to enhance the interpretability of predictions, making the model more transparent and useful for financial analysts and investors.

2.5 APPLICATIONS

The proposed financial time series forecasting model leverages deep learning techniques to enhance various aspects of finance and investment. It aids in stock market prediction by helping traders make informed decisions and identifying trends for better risk management. In portfolio management, it optimizes asset allocation and balances risk and return through accurate price forecasts. The model also enhances algorithmic trading by automating buy/sell decisions and capturing both sequential and non-sequential financial patterns. Additionally, it supports risk assessment by providing insights into market volatility and financial stability. With the help of Explainable AI (XAI), it assists financial analysts in understanding market movements and ensures transparency in decision-making. Moreover, it contributes to economic and policy research by helping policymakers analyze financial trends and monitor market stability.

CHAPTER 3: LITERATURE REVIEW

3.1 INTRODUCTION

Stock price prediction has been a significant area of research, with deep learning (DL) models playing a crucial role in advancing forecasting accuracy. Financial markets are highly volatile, and traditional machine learning (ML) methods often struggle to capture complex temporal dependencies. Deep learning, particularly models like Long Short-Term Memory (LSTM) and hybrid architectures, has shown promise in improving predictive performance.

To establish a strong foundation for our research, we conducted a systematic literature review focusing on stock price prediction using deep learning. Our search began with Google Scholar, using specific keywords such as ‘stock price prediction’, ‘deep learning’, ‘publishing years between 2020-2024’, to retrieve relevant studies. Given the abundance of available literature, we filtered papers based on credibility, considering only those published in renowned journals and conference proceedings from publishers such as Springer, Elsevier, and arXiv. We excluded non-peer-reviewed sources and studies that lacked empirical validation on real-world datasets.

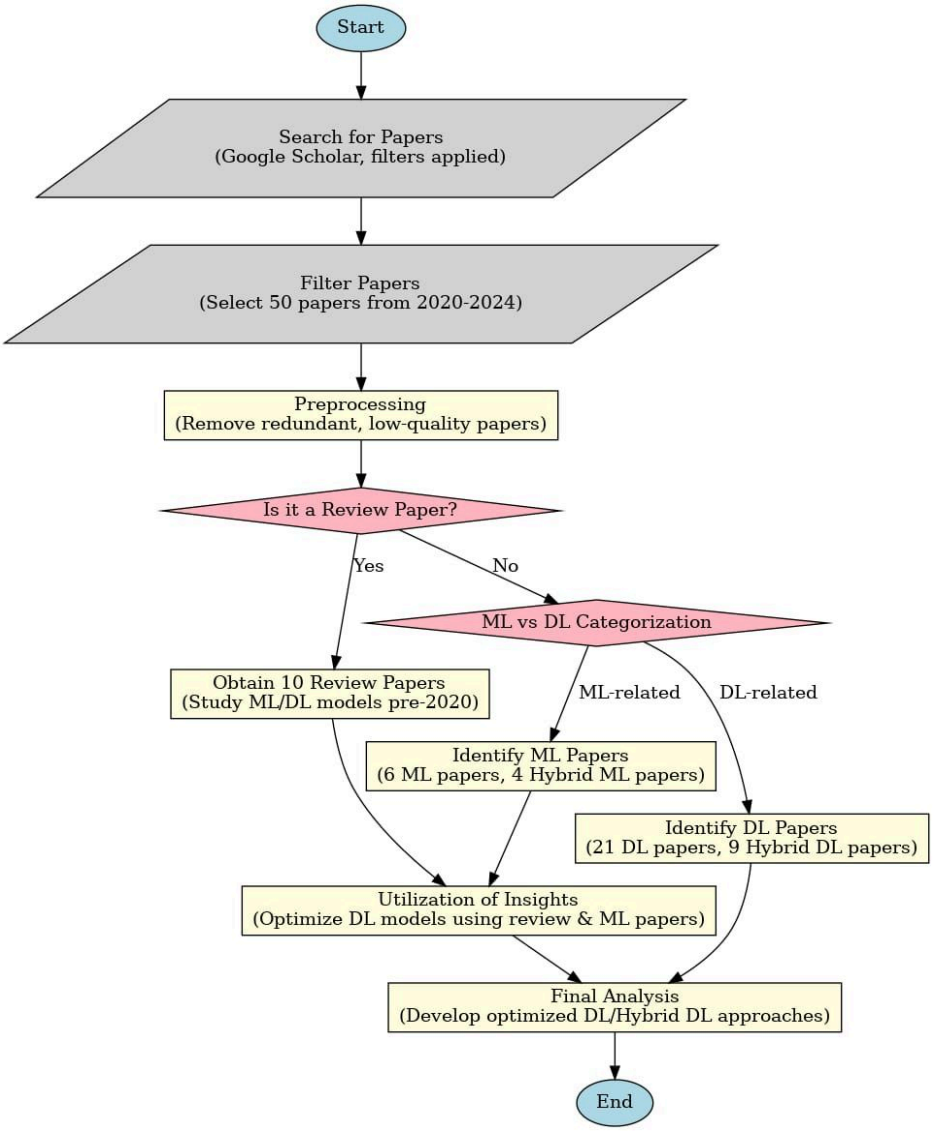


Figure 1: Flowchart for research papers selection process

Additionally, we analyzed the geographical distribution of selected research papers to understand global research trends in financial forecasting. The country-wise distribution graph (Fig. 2) illustrates that a significant number of studies originate from the USA, China, and India, indicating the global relevance of stock market forecasting. This distribution reflects the interest and advancements in financial AI research across different economic regions.

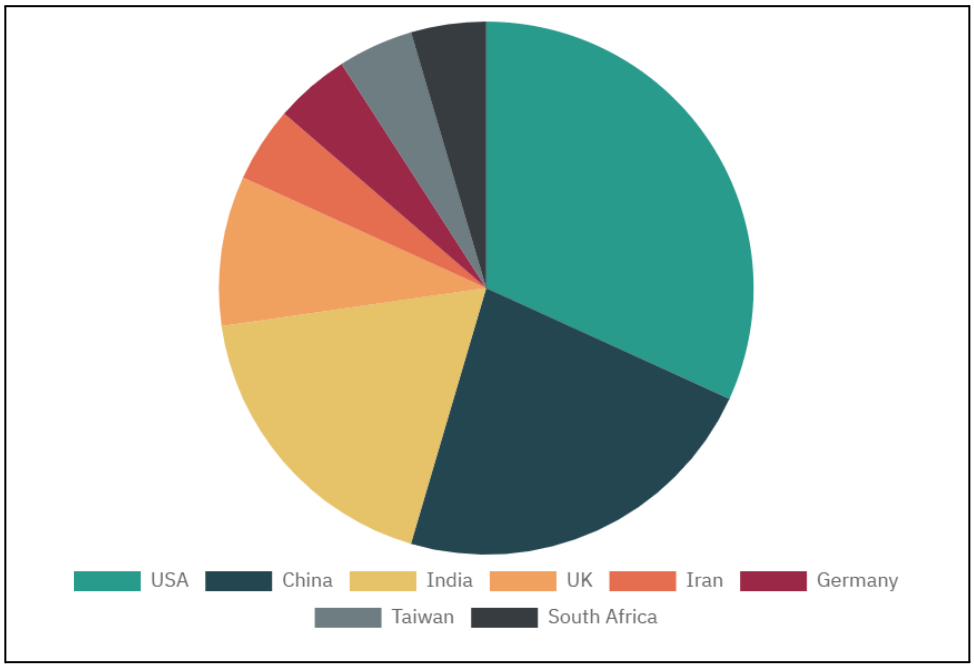


Figure 2: Country-wise distribution of the collected research papers

By narrowing our focus to deep learning and hybrid models, we aimed to explore the most effective techniques for stock price prediction. The subsequent sections of this review delve deeper into these studies, comparing methodologies, datasets, and key findings to identify existing research gaps and potential areas for improvement.

3.2 TRADITIONAL FORECASTING MODELS

Financial time series forecasting has long relied on traditional statistical models known for their interpretability and established theoretical foundations. Among these, the Autoregressive Integrated Moving Average (ARIMA) model has remained a cornerstone in financial prediction tasks. ARIMA effectively captures linear dependencies in time series data by combining autoregression, differencing, and moving average components. Several studies have benchmarked ARIMA's effectiveness, particularly for short-term forecasts. For instance, a study by *Gajamannage et al. ([13/03/2023], Elsevier: Expert Systems with Applications)* demonstrated that while ARIMA efficiently captures linear trends, it struggles with complex, non-linear patterns inherent in financial markets[5].

The GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model has also been widely employed for modeling volatility in financial time series data. Its strength lies in effectively representing time-varying volatility, making it suitable for risk management applications. GARCH models effectively capture volatility clustering in stock price movements but may fail to incorporate long-term dependencies without additional enhancements.

In addition, traditional machine learning algorithms like Support Vector Machines (SVM) and Random Forest (RF) have been employed to complement statistical models. While these methods introduce non-linear mapping capabilities, they often require extensive feature engineering to achieve optimal performance. As discussed in the study by *Zhang et al. ([31/10/2022], Journal of Industrial Engineering and Applied Science)*, these models can be effective for short-term forecasts but may underperform when dealing with sharp price fluctuations[3].

Overall, while traditional models like ARIMA and GARCH have laid the groundwork for financial forecasting, their limited capacity to capture intricate, dynamic patterns has necessitated the exploration of more advanced machine learning and deep learning techniques.

3.3 MACHINE-LEARNING BASED MODELS

Machine learning models have emerged as powerful tools for financial time series forecasting due to their ability to capture complex, non-linear patterns. Among the prominent techniques, LightGBM and CatBoost have gained popularity for their efficiency in handling large-scale data and robust performance. *Srivastava et al. ([01/03/2024], Elsevier: Applied Soft Computing)* demonstrated that a hybrid approach combining association rule mining, linear regression, and LightGBM achieved improved predictive accuracy by leveraging both data-driven insights and algorithmic adaptabilit.[2].

Furthermore, Gated Recurrent Units (GRU) have shown promise in modeling sequential data efficiently. In their study, *Srivastava et al. ([01/03/2024], Elsevier: Applied Soft Computing)* highlighted GRU's ability to reduce computational overhead while maintaining performance comparable to more complex architectures like LSTM[2].

Moreover, hybrid models integrating machine learning methods with traditional techniques have gained traction. *Zhao et al. ([31/01/2024], Elsevier: Applied Soft Computing)* introduced a GridSearchWEF model that combines weighted error functions with machine learning-based grid search techniques, improving both accuracy and stability in financial forecasting tasks[1].

While machine learning models enhance predictive performance through data-driven learning, their reliance on extensive training data and hyperparameter tuning remains a notable challenge. Consequently, researchers have increasingly explored hybrid frameworks that combine machine learning capabilities with deep learning architectures for improved robustness.

3.4 DEEP LEARNING-BASED MODELS

Deep learning models have demonstrated remarkable potential in financial time series forecasting by effectively capturing complex temporal dependencies. Among these, the Long Short-Term Memory (LSTM) network has been extensively utilized due to its superior ability to manage sequential data and retain long-term dependencies. As noted by *Fang et al. ([31/10/2022], Elsevier: Expert Systems with Applications)*, LSTM models equipped with batch normalization techniques have shown improved stability and robustness in volatile financial markets. LSTM models have emerged as the most widely explored deep learning architecture for financial forecasting, dominating recent research trends[4].

The Recurrent Neural Network (RNN) has also been a popular choice, demonstrating its strength in capturing sequential dependencies despite its susceptibility to vanishing gradient issues. Gated Recurrent Units (GRU) and Convolutional Neural Networks (CNN) have gained traction for their improved efficiency and feature extraction capabilities.

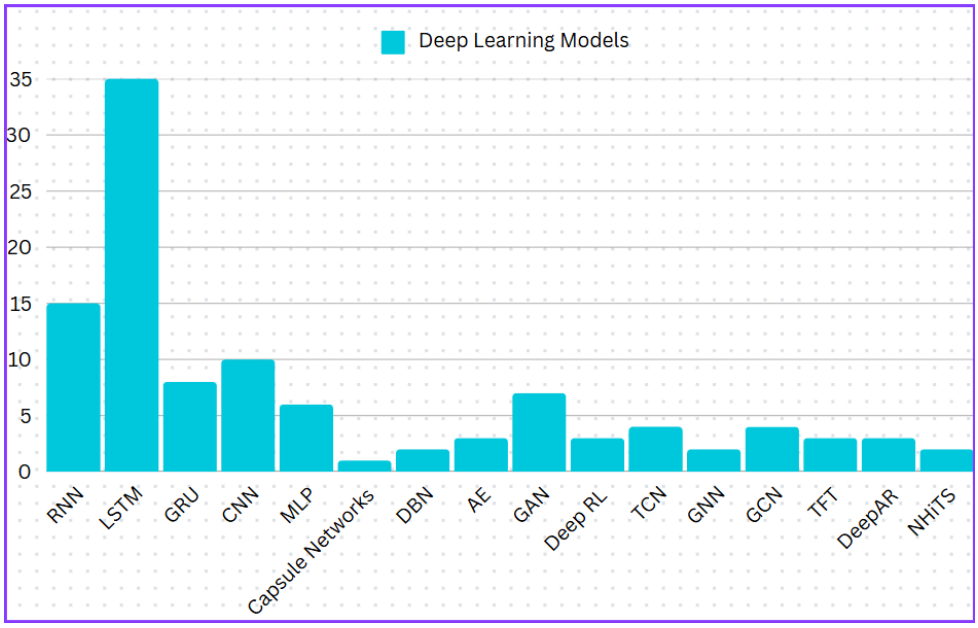


Figure 3: Distribution of deep learning models used in collected research papers

More specialized architectures like Capsule Networks, DBN, AE, and Deep RL have appeared less frequently in research but are gradually gaining attention for their unique strengths in handling complex data patterns. Notably, Generative Adversarial Networks (GANs) have shown promise in data augmentation and improving prediction robustness.

The Bidirectional LSTM (Bi-LSTM) architecture has further enhanced predictive performance by processing data in both forward and backward directions. *Zhang et al. ([01/08/2024], JIEAS: Journal Of Industrial Engineering and Applications)* reported that Bi-LSTM achieved superior accuracy when compared to traditional LSTM and other baseline models, especially in capturing intricate dependencies within stock price patterns[3]. The emerging Dual-LSTM model, introduced by *Gajamannage et al. ([13/03/2023], Elsevier: Expert Systems with Applications)*, demonstrated improved real-time forecasting capabilities by leveraging two interconnected LSTM layers to enhance feature extraction and reduce prediction errors[5].

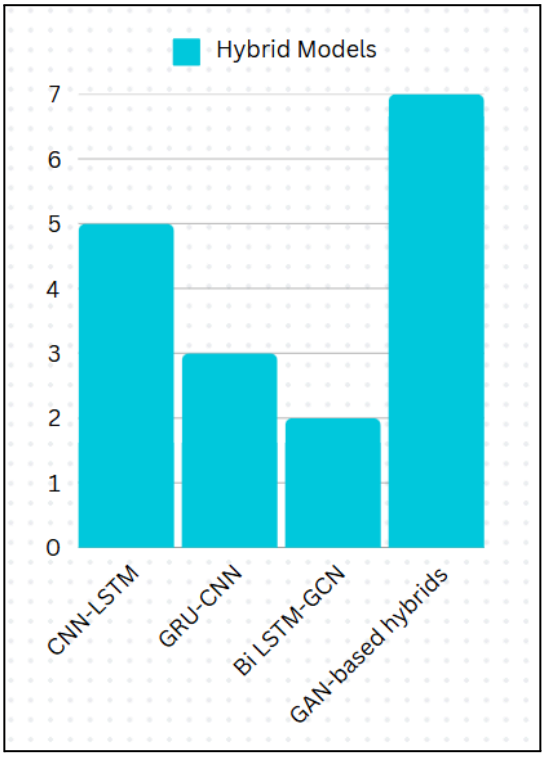


Figure 4: Distribution of hybrid Deep Learning models used in collected research papers

Hybrid models combining deep learning with traditional statistical techniques have gained attention. For instance, *Zhao et al. ([31/01/2024], Elsevier: Applied Soft Computing)* introduced a model that integrates weighted error functions with LSTM to optimize accuracy in financial forecasts. Notable hybrid architectures such as CNN-LSTM and GRU-CNN have demonstrated strong results. Furthermore, advanced combinations like GAN-based hybrids have emerged as a leading hybrid framework, reflecting the growing preference for such versatile models in capturing intricate financial patterns[1].

While deep learning architectures excel in complex pattern recognition, their computational complexity and risk of overfitting highlight the importance of appropriate model design and hyperparameter tuning strategies for optimal performance.

3.5 HYBRID MODELS AND EMERGING TECHNIQUES

In recent years, hybrid models have gained significant attention in financial forecasting due to their ability to combine the strengths of different architectures. By integrating various models, hybrid approaches aim to enhance predictive accuracy and address limitations present in individual models.

For instance, CNN-LSTM models have proven effective in capturing both spatial and temporal patterns, allowing improved feature extraction from financial data. GRU-CNN hybrids have also been explored, showcasing improved performance in handling non-linearities inherent in stock price movements. Similarly, Bi-LSTM-GCN frameworks have demonstrated strong results in complex graph-based dependencies, further improving prediction accuracy. GAN-based hybrid models have emerged as a prominent technique, effectively leveraging adversarial learning to enhance data generation and improve robustness in volatile financial markets.

As reported by *Zhao et al. ([31/01/2024], Elsevier: Applied Soft Computing)*, hybrid models integrating weighted error functions with LSTM have demonstrated improved accuracy by minimizing forecasting errors. Moreover, integrating GANs with traditional deep learning frameworks has shown notable improvements in predicting extreme market fluctuations, enhancing the robustness of financial prediction systems[1].

3.6 EXPLAINABLE AI (XAI) IN FINANCIAL FORECASTING

With the increasing complexity of deep learning models in financial forecasting, the need for transparency and interpretability has become crucial. Explainable AI (XAI) techniques have emerged as essential tools for improving model trustworthiness and ensuring financial decision-makers can understand the rationale behind predictions.

SHAP (SHapley Additive Explanations) has been widely employed to quantify the contribution of individual features to model outputs, providing valuable insights into market-driving factors.

Similarly, LIME (Local Interpretable Model-agnostic Explanations) has proven effective in analyzing complex financial data by generating locally interpretable models that explain individual predictions. By utilizing LIME, financial analysts can gain insights into model behavior under various market conditions, improving confidence in model predictions.

In addition to these established methods, attention mechanisms have emerged as powerful tools in enhancing interpretability. Attention-based models allocate varying importance to different input features, providing insights into which data points influence predictions most significantly. As noted by *M.M Billah et al. ([20/09/2023], Springer: Neural Computing and Applications)*, attention mechanisms have demonstrated strong performance in improving the explainability of LSTM and Transformer-based models[30].

The integration of XAI techniques is increasingly recognized as essential in ensuring the practical applicability of advanced forecasting models, fostering trust in automated financial systems, and improving decision-making in volatile financial environments.

3.7 KEY DATASETS AND BENCHMARKING

Accurate and comprehensive datasets are fundamental in evaluating financial forecasting models, ensuring their reliability and applicability in real-world scenarios. Researchers have employed various publicly available and proprietary datasets to benchmark model performance effectively.

The Yahoo Finance dataset has been extensively used due to its wide availability and comprehensive historical market data, including stock prices, indices, and other financial metrics. This dataset has served as a reliable benchmark for evaluating LSTM and GRU-based models in volatile market conditions.

The Quandl dataset has been widely employed for evaluating deep learning frameworks. As noted by *Zhang et al. ([05/09/2023], Springer: Neural Computing and Applications)*, Quandl's extensive economic and financial data repository has enabled the testing of models such as GAN-based hybrids and Transformer models in capturing intricate financial patterns[21].

In addition, proprietary datasets from major stock exchanges like the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE) have provided valuable insights into regional market trends. Research by *Gajamannage et al. ([13/03/2023], Elsevier: Expert Systems with Applications)* utilized BSE data to demonstrate the effectiveness of Dual-LSTM models in real-time financial forecasting[5].

Benchmarking practices have evolved to include various performance metrics, with Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) being the most commonly employed. As reported by *Zhao et al. ([31/01/2024], Elsevier: Applied Soft Computing)*, RMSE has been particularly effective in assessing model stability in volatile financial environments. Furthermore, researchers have increasingly emphasized the use of directional accuracy metrics to evaluate the practical applicability of models in stock price movement prediction[1].

The integration of diverse datasets, along with robust benchmarking frameworks, remains crucial in advancing financial forecasting research and ensuring model performance is effectively validated in real-world conditions.

CHAPTER 4: PROBLEM IDENTIFICATION AND OBJECTIVES

This chapter outlines the research problem, challenges, and objectives of this study on stock price forecasting using deep learning techniques. The study aims to assess the performance of a hybrid Bi-LSTM + Dense Network model in predicting stock prices in the Indian financial market, specifically using BSE SENSEX NEXT50 (2014–2024) data. Additionally, Explainable AI (XAI) is incorporated to enhance interpretability, ensuring that predictions are more transparent and meaningful for financial analysts and investors.

4.1 PROBLEM STATEMENT

Stock price prediction remains a complex challenge due to the inherent volatility and non-linearity of financial markets. Traditional forecasting models, such as statistical and machine learning techniques, struggle to capture intricate patterns in financial time series data, leading to unreliable predictions. While deep learning models have shown promise in improving accuracy, their applicability in the Indian market remains underexplored.

Furthermore, most deep learning models operate as black boxes, making it difficult for investors and analysts to interpret predictions and trust model outputs. The lack of explainability and interpretability in AI-driven forecasting limits its practical adoption in financial decision-making.

This project seeks to evaluate the effectiveness of deep learning techniques, specifically a hybrid Bi-LSTM + Dense Network model, in forecasting stock prices within the Indian financial market. By using historical data from the BSE SENSEX NEXT50 index (2014–2024), the study examines whether deep learning can effectively model sequential dependencies and improve predictive accuracy. Additionally, integrating Explainable AI (XAI) ensures transparency and enhances trust in model-driven predictions.

4.2 CHALLENGES AND GAPS IN EXISTING RESEARCH

Despite extensive research on stock price forecasting, several key challenges and gaps persist in existing studies:

1. High Market Volatility & Non-Stationary Data

Financial markets are influenced by multiple factors such as macroeconomic conditions, geopolitical events, and investor sentiment, making price movements highly volatile and unpredictable. Many traditional models fail to adapt dynamically to these changing conditions, reducing forecasting accuracy.

2. Limitations of Traditional Statistical & ML Models

- Classical time series models like ARIMA, GARCH, and VAR often fail to capture long-term dependencies and non-linear relationships in financial data.
- Traditional machine learning models like Random Forest, SVM, and XGBoost rely on handcrafted feature selection and struggle with sequential data representation.

3. Need for More Advanced Deep Learning Models

Deep learning models such as LSTMs and CNNs have demonstrated better performance in financial forecasting. However, there is limited research on hybrid deep learning architectures (e.g., Bi-LSTM + Dense Network) in the Indian stock market, and their effectiveness in capturing both short-term and long-term dependencies needs further exploration.

4. Lack of Explainability in AI Models

- Many existing deep learning models function as black boxes, making it difficult for financial analysts to trust and validate their predictions.
- The absence of Explainable AI (XAI) techniques prevents practical adoption in the investment sector, where decisions require high accountability and transparency.

5. Underrepresentation of Indian Market Studies

- Most research on deep learning for stock price prediction has been conducted in Western markets (NASDAQ, S&P 500, NYSE, etc.), while studies focusing on Indian financial markets remain limited.
- The unique market dynamics of India, such as policy-driven fluctuations, regulatory interventions, and retail investor participation, necessitate models tailored to this financial landscape.

This project addresses these gaps by implementing and analyzing a hybrid Bi-LSTM + Dense Network model on Indian stock market data and integrating Explainable AI to improve transparency.

CHAPTER 5: EXISTING SYSTEM AND PROPOSED SYSTEM

5.1 BASELINE MODELS

5.1.1 ARIMA Model

The AutoRegressive Integrated Moving Average (ARIMA) model is a widely used statistical technique for time series forecasting. It combines autoregressive (AR), differencing (I), and moving average (MA) components to model temporal dependencies and account for trends in time series data.

ARIMA consists of three key components:

1. Autoregressive (AR) Component – Uses past values to predict future values.
2. Integrated (I) Component – Differencing is applied to make the time series stationary.
3. Moving Average (MA) Component – Uses past forecast errors to improve predictions.

Parameters:

ARIMA is denoted as ARIMA(p, d, q), where:

- p – Number of lag observations in the AR component.
- d – Number of times differencing is applied to achieve stationarity.
- q – Number of lagged forecast errors in the MA component.

Equation

The general ARIMA equation is: $\Phi(L)(1-L)^{d}Y_t=c+\Theta(L)\epsilon_t$

Where,

- $\Phi(L)=1-\phi_1L-\phi_2L^2-\cdots-\phi_pL^p$ (Autoregressive part)
- $\Theta(L)=1+\theta_1L+\theta_2L^2+\cdots+\theta_qL^q$ (Moving Average part)
- $(1-L)^d$ represents differencing of order d
- c is a constant term, and ϵ_t is the error term

Figure 5: Block diagram of ARIMA model building process (*Source*: *SN Computer Science, "Performance Evaluation of Soft Computing Approaches for Forecasting COVID-19 Pandemic Cases," Fig. 2, [2021]*)

ARIMA is effective for stationary time series and is commonly used in financial forecasting, stock market predictions, and economic modeling.

5.1.2 SARIMA Model

SARIMA (Seasonal ARIMA) is an extension of the ARIMA model that incorporates seasonality. It is used for time series forecasting when data exhibits repeating patterns at fixed intervals. SARIMA models both short-term dependencies and seasonal effects to improve prediction accuracy.

SARIMA is defined by the parameters (p, d, q) × (P, D, Q, m), where:

- (p, d, q): ARIMA components
  - p = Number of autoregressive (AR) terms
  - d = Degree of differencing
  - q = Number of moving average (MA) terms
- (P, D, Q, m): Seasonal components
  - P = Seasonal autoregressive order
  - D = Seasonal differencing order
  - Q = Seasonal moving average order
  - m = Seasonal period (e.g., 12 for monthly data, 4 for quarterly data)

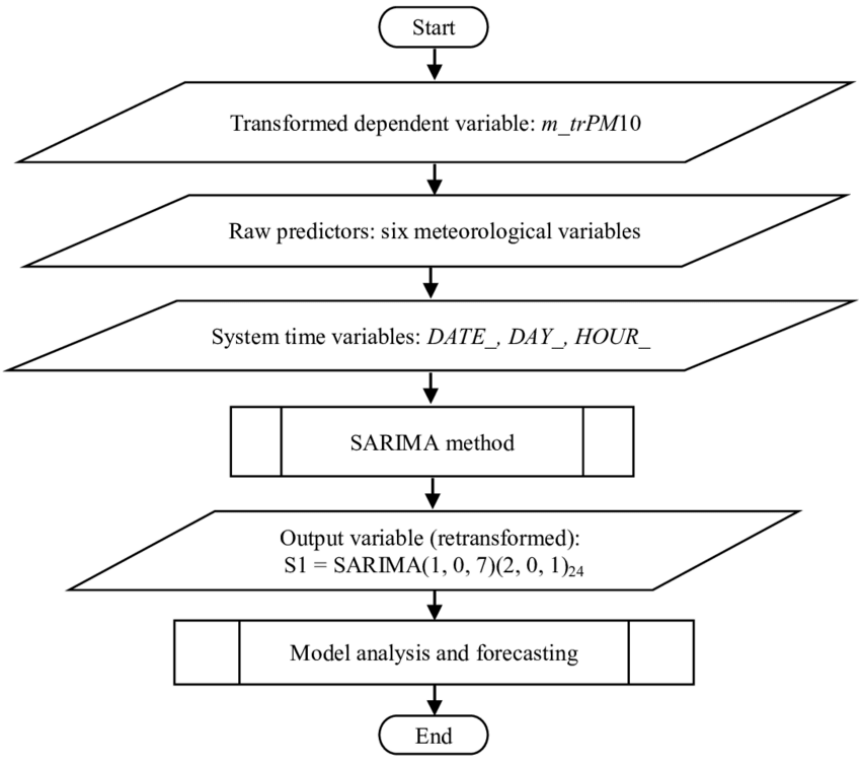


Figure 6: Working of SARIMA model (*Source: S. Gocheva-Ilieva, "Assaying SARIMA and Generalised Regularised Regression for Particulate Matter PM10 Modelling and Forecasting," Fig. 3, [2019]*)

The SARIMA model is expressed as:

$$\Phi_P(L^m)\Phi_p(L)(1-L)^d(1-L^m)^D Y_t = c + \Theta_Q(L^m)\Theta_q(L)\epsilon_t$$

Where:

- $\Phi_p(L)$ and $\Phi_P(L^m)$ represent the non-seasonal and seasonal autoregressive terms.
- $(1-L)^d$ and $(1-L^m)^D$ perform non-seasonal and seasonal differencing.
- $\Theta_q(L)$ and $\Theta_Q(L^m)$ represent the non-seasonal and seasonal moving average terms.
- ϵ_t is the white noise error term.

SARIMA effectively captures both trend and seasonality, making it suitable for financial time series, retail sales, and climate data forecasting.

5.1.3 Convolutional Neural Network (CNN) - 1D,2D,3D

A Convolutional Neural Network (CNN) is a deep learning model designed for feature extraction and pattern recognition in structured data. It uses convolutional layers to detect spatial and temporal dependencies, making it suitable for image, sequence, and volumetric data processing.

CNN consists of the following key layers:

1. Convolutional Layer – Applies filters to input data, extracting meaningful features.
2. Activation Function (ReLU) – Introduces non-linearity to the model.
3. Pooling Layer – Reduces dimensionality while retaining important features (e.g., max pooling).
4. Fully Connected Layer – Connects extracted features for final classification or regression.

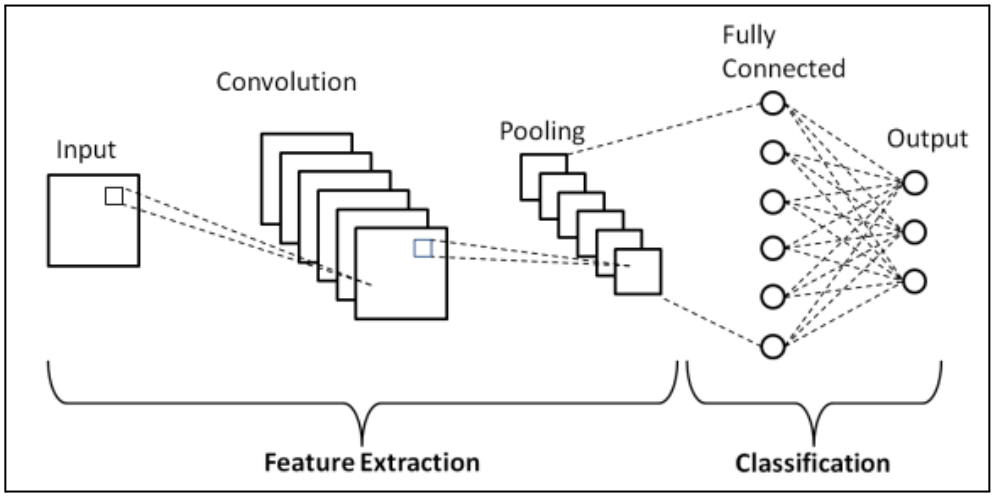


Figure 7: Architecture of CNN model (*Source: https://www.upgrad.com/blog/basic-cnn-architecture/*)

Types of CNN Models:

- CNN-1D (One-Dimensional)
  i. Used for sequential data like time series and speech signals.
  ii. Convolution operates along a single axis (temporal/spatial).
  iii. Mathematical Equation: $y_j = f(\sum w_i \cdot x_{j+i} + b)$

  w_i = filter weights, x_{j+i} = input, b = bias, f = activation function.

- CNN-2D (Two-Dimensional)
  i. Used for image processing (e.g., classification, object detection).
  ii. Convolution operates over height and width dimensions.
  iii. Mathematical Equation: $y_{i,j} = f(\sum\sum w_{m,n} \cdot x_{i+m,j+n} + b)$

  $w_{m,n}$ = filter weights, $x_{i+m,j+n}$= input, b = bias.

- CNN-3D (Three-Dimensional)
  i. Used for volumetric data like medical imaging and video processing.
  ii. Convolution operates over three dimensions (height, width, depth).
  iii. Mathematical Equation: $y_{i,j,k} = f(\sum\sum\sum w_{m,n,o} \cdot x_{i+m,j+n,k+o} + b)$

  O = depth of the filter.

The parameters are as follows:

- Kernel size: Defines the size of the filter.
- Stride: Controls the step size during convolution.
- Padding: Maintains spatial dimensions after convolution.
- Number of filters: Determines feature extraction capacity.

CNNs are widely used in image classification, time series forecasting, and video analysis due to their ability to learn spatial hierarchies of features.

5.1.4 Recurrent Neural Network

A Recurrent Neural Network (RNN) is a deep learning model designed to process sequential data by maintaining a memory of past inputs. It is widely used for time series forecasting, natural language processing (NLP), and speech recognition.

Architecture & Working:

1. Input Layer – Takes sequential data as input.
2. Hidden Layer (Recurrent) – Stores past information using hidden states, enabling temporal dependencies.
3. Output Layer – Generates predictions based on the current input and past memory.
4. Feedback Connection – Unlike traditional neural networks, RNNs have loops to retain past information.

At each time step t, the hidden state h_t is updated using the previous hidden state h_{t-1} and the current input x_t.

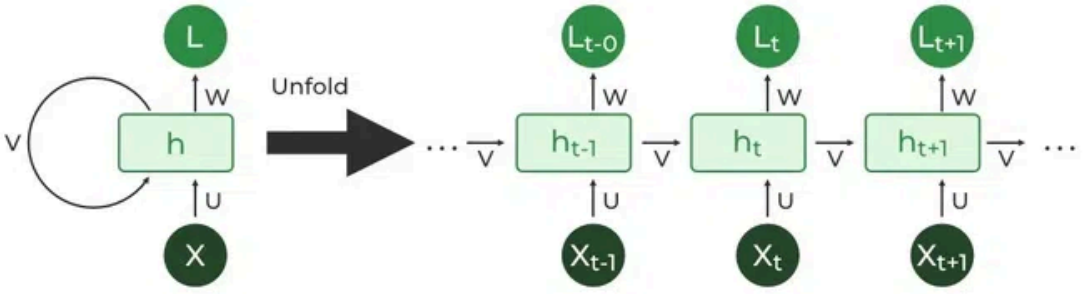


Figure 8: Architecture of RNN model (*Source: https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/*)

Mathematical Equation:

$$h_t=f(W_h h_{t-1}+W_x x_t+b)$$

$$y_t=g(W_y h_t+c)$$

Where:

- h_t = Hidden state at time t.
- x_t = Input at time t.
- W_h, W_x, W_y= Weight matrices.
- b,c= Bias terms.
- f = Activation function (typically tanh or ReLU).
- g = Output activation function (softmax for classification, linear for regression).

Parameters:

- Hidden units – Number of neurons in the recurrent layer.
- Learning rate – Controls weight updates.
- Sequence length – Defines how far back dependencies are considered.
- Activation function – Tanh, ReLU, or sigmoid for non-linearity.

RNNs struggle with long-term dependencies due to vanishing gradient issues, which are addressed using advanced variants like LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit).

5.1.5 Long Short Term Memory (LSTM) Model

LSTM is a type of Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem. It efficiently captures long-term dependencies in sequential data, making it ideal for time series forecasting, natural language processing, and speech recognition.

Architecture & Working:

LSTM introduces memory cells with gates that regulate information flow:

1. Forget Gate (F_t) – Decides what past information to discard.
2. Input Gate (I_t) – Determines what new information to store in the cell state.
3. Cell State (C_t) – Stores long-term memory.
4. Output Gate (O_t) – Controls the final output from the memory cell.

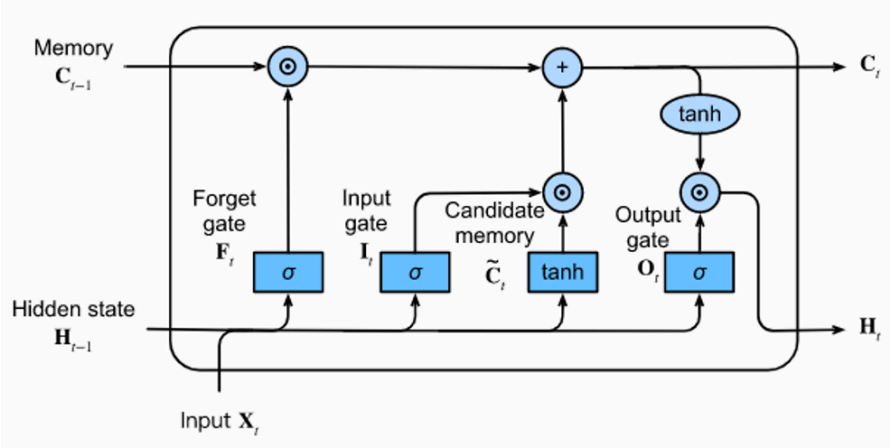


Figure 9: Architecture of LSTM model (*Source: https://d2l.ai/chapter_recurrent-modern/lstm.html*)

At each time step t, the LSTM cell updates using the following equations:

Mathematical Equations:

$$f_t=\sigma(W_f \cdot [h_{t-1},x_t]+b_f)$$

$$i_t=\sigma(W_i \cdot [h_{t-1},x_t]+b_i)$$

$$\bar{C}_t=\tanh(W_C \cdot [h_{t-1},x_t]+b_C)$$

$$C_t=f_t \cdot C_{t-1}+i_t \cdot \bar{C}_t$$

$$o_t=\sigma(W_o \cdot [h_{t-1},x_t]+b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

Where:

- x_t = Input at time t.
- h_t = Hidden state at time t.
- C_t = Cell state at time t.
- W_f, W_i, W_C, W_o = Weight matrices.
- b_f, b_i, b_C, b_o = Bias terms.
- σ = Sigmoid activation function.
- tanh = Hyperbolic tangent activation function.

Parameters:

- Hidden units – Number of neurons in each LSTM cell.
- Learning rate – Controls weight updates.
- Sequence length – Defines how many past time steps are considered.
- Activation functions – Sigmoid for gating mechanisms, Tanh for state updates.

LSTM models are widely used for financial forecasting, where capturing long-term dependencies in stock price trends is crucial.

5.1.6 Gated Recurrent Unit (GRU) Model

Gated Recurrent Unit (GRU) is a variant of Recurrent Neural Networks (RNNs) designed to handle sequential data while reducing computational complexity compared to Long Short-Term Memory (LSTM). It achieves this by using fewer gates, making it more efficient while still capturing long-term dependencies.

Architecture & Working:

GRU consists of two main gates:

1. Reset Gate (r_t) – Controls how much past information should be forgotten.
2. Update Gate (z_t) – Determines how much of the past hidden state should be carried forward.

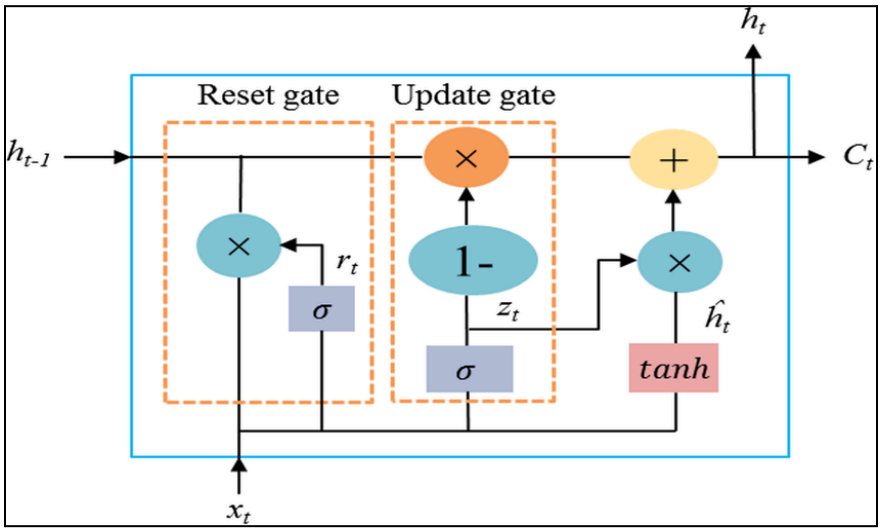


Figure 10: Architecture of GRU model (*Source: Saeed Mohsin, "Recognition of Human Activity Using GRU Deep Learning Algorithm," Multimedia Tools and Applications, Fig. 2, [2023]*)

At each time step t, the GRU updates using the following equations:

Mathematical Equations:

$$r_t=\sigma(W_r \cdot [h_{t-1},x_t]+b_r)$$

$$z_t=\sigma(W_z \cdot [h_{t-1},x_t]+b_z)$$

$$\hat{h}_t=\tanh(W_h \cdot [r_t \odot h_{t-1},x_t]+b_h)$$

$$h_t=(1-z_t)\odot h_{t-1}+z_t \odot \hat{h}_t$$

Where:

- x_t = Input at time t.
- h_t = Hidden state at time t.
- W_r, W_z, W_h = Weight matrices.
- b_r, b_z, b_h = Bias terms.
- σ = Sigmoid activation function.
- tanh = Hyperbolic tangent activation function.
- $\odot$ = Element-wise multiplication.

Parameters:

- Hidden units – Number of neurons in each GRU cell.
- Learning rate – Controls weight updates.
- Sequence length – Defines how many past time steps are considered.
- Activation functions – Sigmoid for gating mechanisms, Tanh for state updates.

GRU is widely used for time series forecasting, including stock price prediction, as it balances performance and efficiency by reducing the number of gates compared to LSTM.

5.1.7 Bi-Directional LSTM (BI-LSTM) Model

Bidirectional Long Short-Term Memory (Bi-LSTM) is an extension of the LSTM model that processes data in both forward and backward directions. This enhances the model's ability to capture dependencies in sequential data by considering past and future contexts simultaneously.

Architecture & Working:

Bi-LSTM consists of two LSTM layers:

1. Forward LSTM – Processes the input sequence from past to future.
2. Backward LSTM – Processes the input sequence from future to past.

The outputs of both LSTMs are concatenated to form the final output.

At each time step t, the hidden state is updated in both directions:

- Forward pass: $\vec{h}_t$
- Backward pass: h_t
- Final output: $h_t=[\vec{h}_t,h_t]$

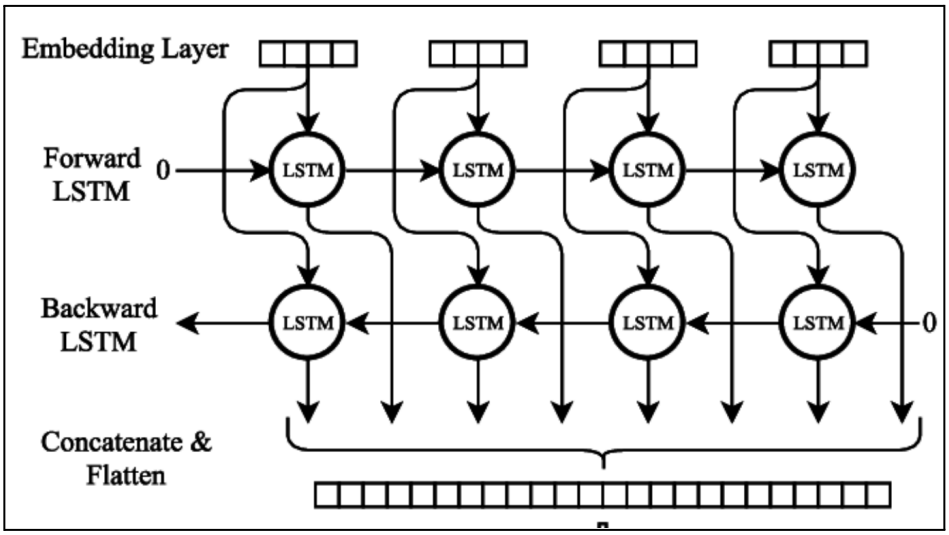


Figure 11: Architecture of Bi LSTM model (*Source: Modelling Radiological Language with Bidirectional Long Short-Term Memory Networks, Cornegruta et al*)

Mathematical Equations:

For the Forward LSTM:

$$\vec{f}_t=\sigma(W_f \cdot [\vec{h}_{t-1},x_t]+b_f)$$

$$\vec{i}_t=\sigma(W_i \cdot [\vec{h}_{t-1},x_t]+b_i)$$

$$\vec{o}_t=\sigma(W_o \cdot [\vec{h}_{t-1},x_t]+b_o)$$

$$\vec{c}_t=\vec{f}_t \odot \vec{c}_{t-1}+\vec{i}_t \odot \tanh(W_c \cdot [\vec{h}_{t-1},x_t]+b_c)$$

$$\vec{h}_t = \vec{o}_t \odot \tanh(\vec{c}_t)$$

For the Backward LSTM, similar equations are applied in reverse order.

The final hidden state:

$$H_t = [\vec{h}_t, h_t]$$

Parameters:

- Number of LSTM units – Determines model complexity.
- Sequence length – Number of time steps considered for training.
- Dropout rate – Prevents overfitting.
- Activation functions – Sigmoid for gates, Tanh for state updates.
- Learning rate – Controls weight updates.

Bi-LSTM is widely used in stock price prediction as it captures both past and future dependencies in financial time series data, improving forecasting accuracy.

5.1.8 Transformer Model

The Transformer model is a deep learning architecture designed for handling sequential data without relying on recurrence. It utilizes a self-attention mechanism to process entire input sequences in parallel, making it highly efficient for time-series forecasting, NLP, and other sequential tasks.

Architecture & Working:

A Transformer consists of an encoder-decoder structure, where:

- Encoder: Processes input data into a contextual representation.
- Decoder: Generates the output sequence based on encoded information.

Each encoder and decoder block contains:

1. Multi-Head Self-Attention – Captures dependencies across the entire sequence.
2. Feedforward Network – Applies non-linearity and learns transformations.
3. Layer Normalization & Residual Connections – Ensures stable training and smooth gradient flow.

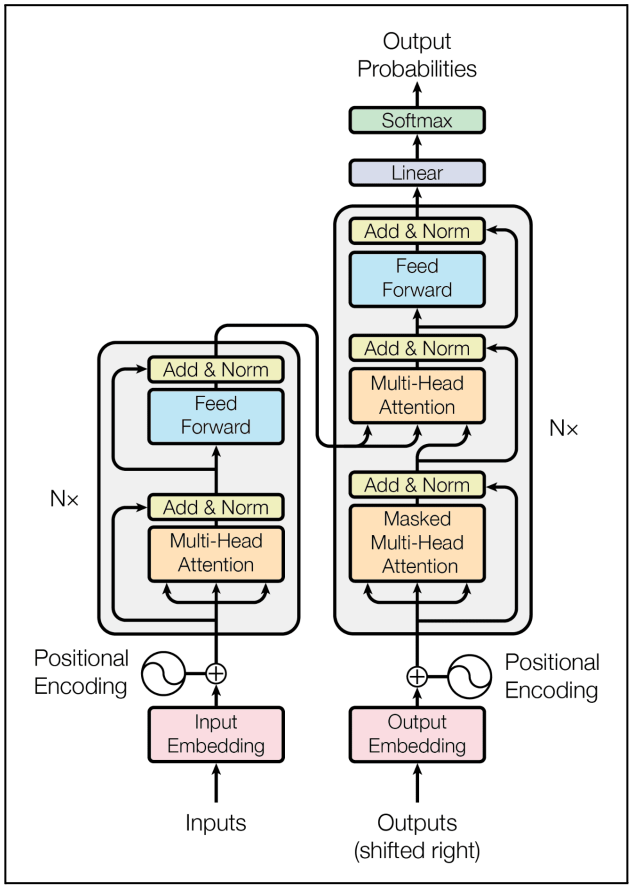


Figure 12: Architecture of Transformer model (*Source: A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention Is All You Need," Advances in Neural Information Processing Systems, Fig. 1, [2017]*)

Self-Attention Mechanism:

Each token attends to all other tokens using a scaled dot-product attention mechanism.

$$\text{Attention}(Q,K,V)=\text{softmax}(QK^T/\sqrt{dk})V$$

Where:

- Q (Query), K (Key), and V (Value) are derived from the input.
- d_k is the dimension of the key vectors.
- The softmax function normalizes attention scores.

Parameters:

- Embedding Dimension (d_{model}) – Defines the size of input and output embeddings.
- Number of Attention Heads (h) – Controls parallel attention mechanisms.
- Number of Layers (N) – Specifies encoder/decoder depth.
- Dropout Rate – Prevents overfitting.
- Learning Rate & Optimizer – Typically uses Adam with warmup scheduling.

The Transformer model is widely used in stock price prediction as it can effectively capture long-range dependencies in financial time series data, making it superior to traditional RNN-based models.

5.1.9 Exponential Smoothing

Exponential Smoothing is a time-series forecasting method that applies exponentially decreasing weights to past observations, giving more importance to recent data while still considering historical values. It is widely used for trend and seasonality modeling in financial markets.

Architecture & Working:

Exponential Smoothing models predict future values by computing a weighted average of past observations, where older values have exponentially decreasing influence. The three main types are:

1. Simple Exponential Smoothing (SES) – Used for data with no trend or seasonality.
2. Holt's Linear Trend Model – Extends SES by incorporating a trend component.
3. Holt-Winters Seasonal Model – Adds a seasonal component to handle periodic patterns.

The general approach involves updating the smoothed value recursively:

$$S_t=\alpha X_t+(1-\alpha)S_{t-1}$$

Where:

- S_t is the smoothed value at time t,
- X_t is the actual observation at time t,
- α (smoothing factor) controls the weight given to recent observations ($0<\alpha<1$).

For Holt's Linear Trend Model, additional equations include:

$$S_t=\alpha X_t+(1-\alpha)(S_{t-1}+b_{t-1})$$

$$b_t=\beta(S_t-S_{t-1})+(1-\beta)b_{t-1}$$

where b_t represents the trend component, and β is the trend smoothing parameter.

For Holt-Winters Seasonal Model, a seasonality factor is added:

$$S_t=\alpha(X_t/L_{t-s}) + (1-\alpha)(S_{t-1}+b_{t-1})$$

$$b_t=\beta(S_t-S_{t-1})+(1-\beta)b_{t-1}$$

$$L_t=\gamma(X_t/S_t) + (1-\gamma)L_{t-s}$$

where L_t represents the seasonal component and γ is the seasonality smoothing parameter.

Parameters:

- α – Smoothing parameter for level.
- β – Smoothing parameter for trend.
- γ – Smoothing parameter for seasonality.
- Seasonal Period (s) – Defines the cycle length for periodic patterns.

Exponential Smoothing is effective for stock price forecasting, especially when trends and seasonal effects are present, making it useful for financial analysts in predicting short-term market movements.

5.1.10 Graph Neural Network

Graph Neural Networks (GNNs) are deep learning models designed to process graph-structured data. Unlike traditional neural networks that work on Euclidean data (e.g., images, text), GNNs operate on nodes, edges, and their relationships, making them suitable for applications like social networks, financial transactions, and stock market prediction.

Architecture & Working:

GNNs learn node representations by iteratively aggregating information from neighboring nodes. The basic process involves:

1. Message Passing – Nodes exchange information with their neighbors.
2. Aggregation – Each node updates its state based on received messages.
3. Update Function – The node's new state is computed using a neural network.
4. Readout Function – The final representation is used for predictions.

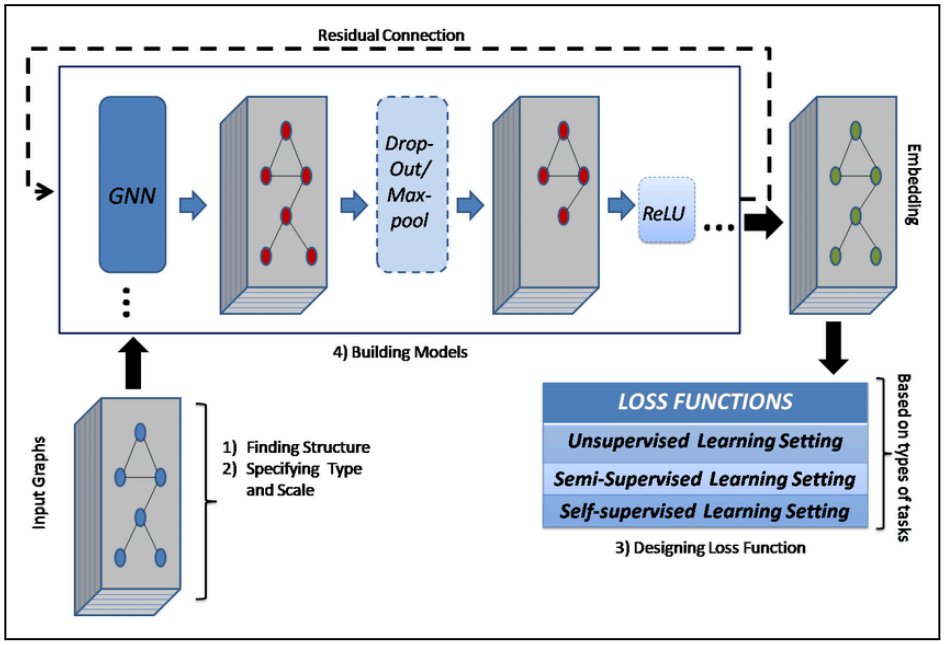


Figure 13: Architecture of GNN model (*Source: L. Waikhom and R. Patgiri, "Graph Neural Networks: Methods, Applications, and Opportunities," 2021, Fig. 3*)

A standard GNN follows the update rule:

$$h_v^{(k)}=\sigma(W^{(k)}\sum(u\in N(v))\ h_u^{(k-1)}+b^{(k)})$$

Where:

- $h_v^{(k)}$ is the node representation at layer k.
- $W^{(k)}$ and $b^{(k)}$ are trainable weights and biases.
- N(v) denotes the neighbors of node v.
- σ is an activation function (e.g., ReLU).

Parameters:

- Node Features (X) – Attributes of nodes (e.g., stock price, volume).
- Edge Features (E) – Relationships between nodes (e.g., correlations between stocks).
- Graph Structure (A) – Adjacency matrix representing connectivity.
- Trainable Weights (W) – Used for transformation and aggregation.

Applications in Stock Prediction:

- Modeling relationships between stocks using financial graphs.
- Capturing complex dependencies in stock price movements.
- Enhancing predictive accuracy by leveraging graph structures.

GNNs are increasingly used in financial markets to analyze stock correlations, detect fraud, and predict asset prices based on relational data.

5.1.11 Residual Network (ResNet) Model

ResNet (Residual Network) is a deep learning architecture designed to solve the vanishing gradient problem in deep neural networks by introducing residual connections (skip connections). It allows for training very deep networks by enabling gradient flow through identity mappings. ResNet is widely used in image processing and time-series forecasting.

Architecture & Working:

ResNet consists of multiple residual blocks, where each block includes a shortcut connection that bypasses one or more layers. The core idea is:

1. Input Transformation: Each layer transforms the input using convolution, batch normalization, and activation functions (ReLU).
2. Residual Connection: The input is directly added to the output of a few layers, ensuring gradients can flow smoothly.

3. Deep Feature Extraction: Multiple residual blocks stacked together extract hierarchical features.
4. Final Classification/Prediction Layer: The processed features are passed through a fully connected layer for prediction.

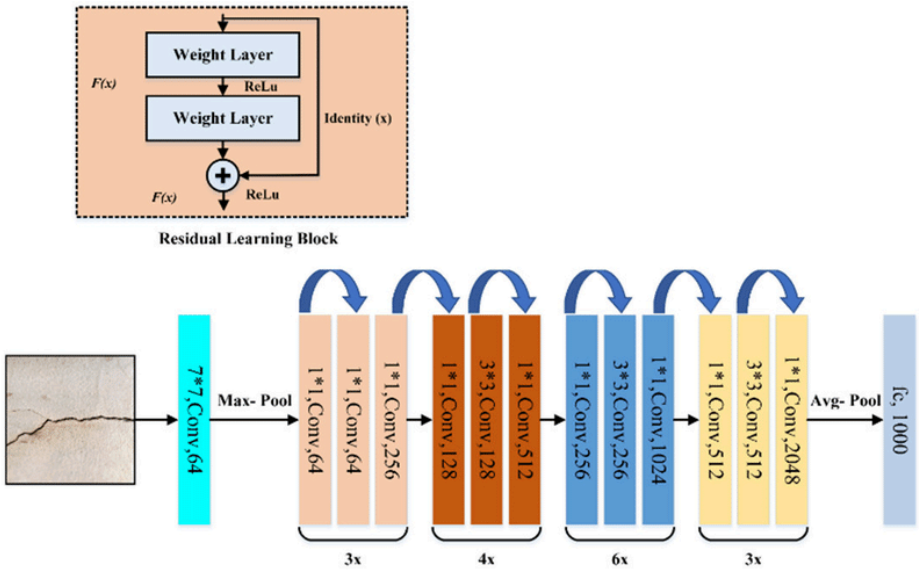


Figure 14: Architecture of ResNet model (*Source: L. Ali and F. S. Alnajjar, "Performance Evaluation of Deep CNN-Based Crack Detection and Localization Techniques for Concrete Structures," 2021, Fig. 6*)

A basic residual block is given by:

$$y=F(x,W)+x$$

Where:

- x is the input,
- F(x,W) is the transformation (convolution, batch normalization, activation),
- W are the learnable weights,
- The output y is the sum of the transformed input and the original input.

Parameters:

- Kernel Size – Defines the size of convolution filters.
- Number of Layers – Determines the depth of the network (e.g., ResNet-18, ResNet-34, ResNet-50, etc.).
- Activation Function – Usually ReLU.
- Batch Normalization – Helps stabilize training.
- Skip Connections – Enable gradient flow and prevent vanishing gradients.

Applications in Stock Prediction:

- Feature extraction from financial time series data.
- Enhancing deep learning models (e.g., CNN-ResNet hybrids).
- Capturing long-term dependencies and patterns.

ResNet's ability to process sequential and structured data efficiently makes it a valuable model in deep learning applications beyond computer vision, including financial forecasting.

5.1.12 GARCH Model

The GARCH model is a statistical approach used to model and predict time-series data where volatility changes over time. It is widely applied in financial markets to estimate stock price volatility by capturing time-dependent heteroskedasticity (changing variance).

Architecture & Working:

GARCH extends the ARCH (Autoregressive Conditional Heteroskedasticity) model by incorporating past variances into the prediction. It models the variance of a time series as a function of its past squared observations and past variances.

1. Mean Equation: The stock return (r_t) is modeled as: $r_t=\mu+\epsilon_t$
   where ϵ_t is a white noise error term.

2. Variance Equation: The conditional variance (σ_t^2) follows: $\sigma_t^2=\omega+\sum\alpha_i\epsilon_{t-i}^2+\sum\beta_j\sigma_{t-j}^2$

   where:

   - ω is a constant
   - α_i are ARCH terms (past squared errors)
   - β_j are GARCH terms (past variances)
   - p and q are the order of the model
3. Iterative Forecasting: Using past observations, the model updates conditional variance estimates over time, capturing volatility clusters in stock prices.

Parameters:

- p – Number of lagged squared error terms (ARCH terms).
- q – Number of past variance terms (GARCH terms).
- ω,α,β – Estimated coefficients that define volatility behavior.
- Conditional Variance (σ_t^2) – Represents predicted volatility at time t.

Applications in Stock Prediction:

- Volatility modeling in financial markets.
- Risk management and portfolio optimization.
- Options pricing and forecasting stock price fluctuations.

GARCH is crucial for understanding market uncertainty and is often used alongside deep learning models for hybrid forecasting approaches.

5.1.13 Prophet Model

The Prophet model is an additive time series forecasting model developed by Facebook, designed to handle missing data, outliers, and seasonal patterns effectively. It is particularly useful for business and financial forecasting, including stock price predictions.

Architecture & Working:

Prophet decomposes time series data into three main components:

1. Trend (g(t)) – Captures the overall growth pattern (linear or logistic).
2. Seasonality (s(t)) – Models recurring patterns (daily, weekly, yearly).
3. Holidays (h(t)) – Adjusts for special events that impact trends.

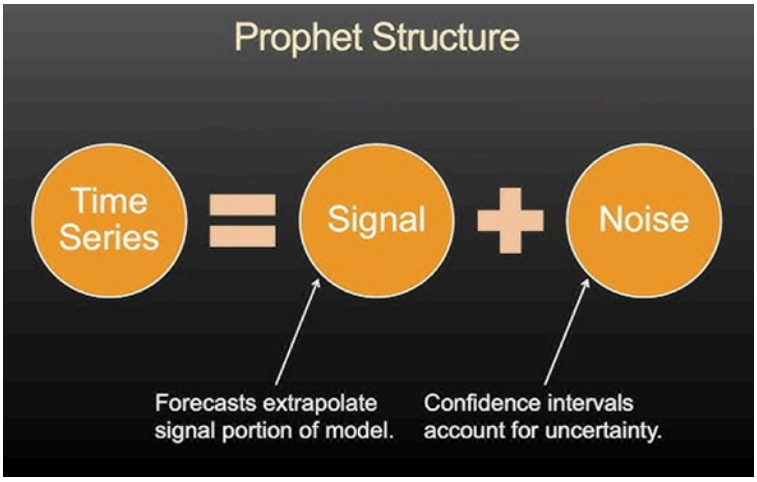


Figure 15: Structure of Prophet model (*Source: E. LaBarr, "What is the Prophet Model," YouTube, 2022*)

The model is expressed as: $y(t)=g(t)+s(t)+h(t)+\epsilon_t$

where ϵ_t represents an error term.

Trend Modeling:

- Linear Growth: $g(t)=kt+m$ where k is the growth rate, and m is the offset.
- Logistic Growth: (when there is a carrying capacity C) $g(t)=C/(1+\exp(-(k(t-m))))$

Seasonality Modeling:

- Modeled using Fourier series to capture periodic patterns.

Parameters:

- k – Growth rate of the trend.
- m – Initial offset of the trend.
- C – Carrying capacity for logistic growth.
- Fourier terms – Control the complexity of seasonal components.
- Holiday effects – Capture the impact of known events.

Applications in Stock Prediction:

- Forecasting stock prices with long-term trends.
- Capturing seasonality in stock movements.
- Handling missing financial data efficiently.

The Prophet model is effective for time series forecasting when deep learning models are too complex, making it useful for stock market analysis.

5.1.14 TimesNet Model

TimesNet is a deep learning-based time-series forecasting model designed to efficiently capture multi-scale temporal patterns. It introduces a novel Time-Series Block (TSBlock), leveraging neural operators to enhance the learning of global dependencies in time-series data.

Architecture & Working:

The architecture of TimesNet is built upon multiple Time-Series Blocks (TimesBlocks) stacked sequentially, each responsible for extracting hierarchical temporal patterns. The key components of the architecture, as illustrated in the provided image, include:

1. Stacked TimesBlocks with Residual Connections:
    - Multiple TimesBlocks are arranged in a stacked manner, allowing the model to learn hierarchical dependencies.
    - Residual connections (+ symbol) ensure stable gradient flow and efficient learning by preserving temporal information across layers.
2. Components of a TimesBlock:
    - Fast Fourier Transform (FFT) for Periods: Converts the time-series data into the frequency domain to extract periodic components.
    - Softmax Layer: Normalizes the extracted frequency components for better representation.
    - Parameter-Efficient Inception Block: Implements multi-scale feature extraction using convolutional operations.
    - Reshape & Reshape Back Modules: Maintain structural consistency between the transformed and original time-series data.
    - Residual Connection: Helps in stabilizing the training process and improving feature retention.

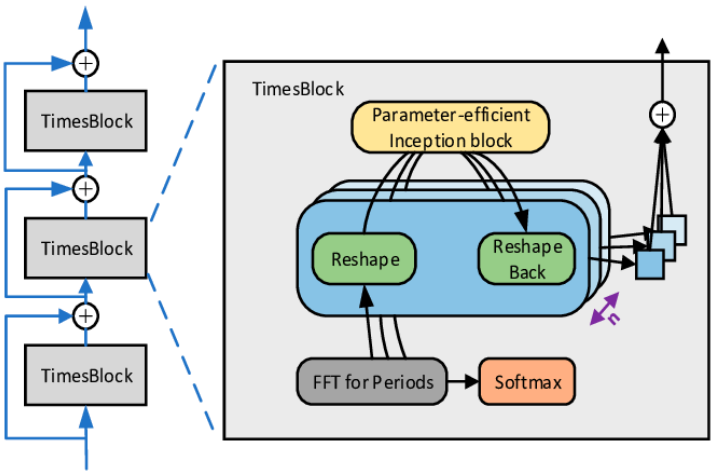


Figure 16: Architecture of TimesNet model (*Source: X. Zhang, K. Yang, and L. Zheng, "Transformer Fault Diagnosis Method Based on TimesNet and Informer," ResearchGate, 2024*)

The overall process starts with embedding the raw time-series data, followed by feature extraction using TimesBlocks. The extracted features are then aggregated using fully connected layers to generate the final predictions.

Mathematical Representation:

TimesNet represents time-series transformations as: $Z=F^{-1}(F(X) \cdot W)$

where:

- X is the input time-series data.
- F and F^{-1} are Fourier Transform and its inverse.
- W represents learnable weights for frequency components.
- Z is the extracted temporal features.

Parameters:

- Window size (T) – Defines the length of the input sequence.
- Hidden dimensions (d) – Controls model complexity.
- Kernel size – Determines receptive field for capturing patterns.
- Dropout rate – Regularization to prevent overfitting.

Applications in Stock Prediction:

- Captures complex time-series dependencies for stock price forecasting.
- Adapts to different temporal patterns, making it robust for financial data.
- Handles non-stationary data using frequency-based transformations.

TimesNet is a state-of-the-art model for time-series forecasting, significantly improving accuracy in financial market predictions.

5.1.15 DeepAR

DeepAR is a probabilistic forecasting model based on deep learning that utilizes autoregressive recurrent neural networks (RNNs) to predict time-series data. It is designed to handle multiple related time-series and can generate probabilistic forecasts by learning patterns across multiple sequences.

Architecture & Working:

DeepAR is built on long short-term memory (LSTM) or gated recurrent unit (GRU) networks and operates in an autoregressive manner. The model takes historical observations and covariates as inputs and predicts future values by sampling from a learned probability distribution. The key components include:

1. Encoder-Decoder Structure – The model first encodes historical observations using an RNN and then decodes future time steps iteratively.
2. Autoregressive Forecasting – Predictions are made step by step, where previous outputs serve as inputs for future time steps.
3. Probabilistic Prediction – Instead of point forecasts, DeepAR predicts probability distributions over future values, commonly modeled using Gaussian or negative binomial distributions.
4. Covariates Integration – The model can incorporate additional external features (e.g., holidays, stock indicators) to improve accuracy.

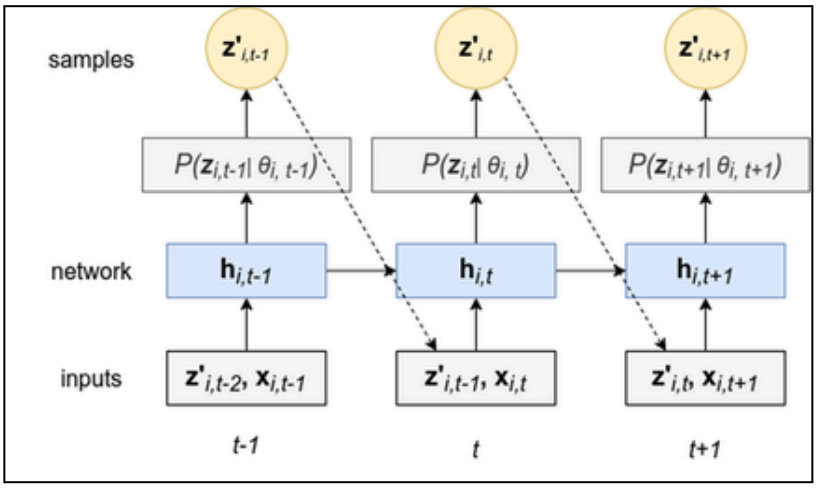


Figure 17: Architecture of DeepAR model (*Source: K. Paramasivan, R. Subburaj, S. Jaiswal, and N. Sudarsanam, "Empirical Evidence of the Impact of Mobility on Property Crimes During the First Two Waves of the COVID-19 Pandemic," ResearchGate, 2022, [Figure 2]*)

Mathematical Representation:

DeepAR models the conditional probability of future values given past observations as:

$$p(X_{t+1:T} | X_{1:t}, Z) = \prod_{(i=t+1,T)} [p(X_i | X_{1:i-1}, Z; \theta)]$$

where:

- $X_{1:t}$ represents historical time-series values.
- Z denotes additional covariates.
- $p(X_i | X_{1:i-1}, Z; \theta)$ is the learned probability distribution at time i.
- θ represents model parameters.

Parameters:

- RNN hidden size (h) – Defines the number of hidden units in the LSTM/GRU layers.
- Number of layers (L) – Determines the depth of the recurrent network.
- Learning rate (α) – Controls model training speed.
- Dropout rate (d) – Regularization to prevent overfitting.
- Likelihood function (p(x)) – Defines the distribution used for probabilistic forecasting (e.g., Gaussian, Negative Binomial).

Applications in Stock Prediction:

- Handles multiple stock time-series simultaneously, improving forecasting accuracy.
- Captures long-term dependencies using RNN-based architecture.
- Provides probabilistic forecasts, helping in risk assessment and decision-making.

DeepAR is widely used for financial forecasting, demand prediction, and anomaly detection due to its ability to model complex time-series dependencies effectively.

5.2 HYBRID MODELS

5.2.1 CNN 3D + TimesNet

Working:

The CNN 3D + TimesNet model is a hybrid deep learning architecture that combines 3D Convolutional Neural Networks (CNNs) for spatial-temporal feature extraction with a fully connected (dense) neural network (TimesNet) to learn complex representations and make predictions.

1. CNN 3D Component:
    a. Spatial-Temporal Feature Extraction: The 3D CNN applies convolutions across multiple frames or time steps to capture both spatial and temporal patterns in the data. This helps the model learn hierarchical spatial representations and localized features from the input data (e.g., stock price data).

    b. Local Pattern Learning: Through the use of 3D convolution, the model is able to extract and learn local patterns across different time steps, which is especially useful for analyzing sequential data like stock prices.

2. TimesNet Component:
    a. Sequential Feature Learning: TimesNet, in this case, is a fully connected (dense) network that processes the features extracted by the CNN. It takes in the original sequential data and applies dense layers to capture complex relationships in the time-series data.
    b. Learning Temporal Dependencies: Though TimesNet is not an LSTM or BI-LSTM, it still helps model dependencies by learning from the sequential data using dense layers, which are effective for capturing non-sequential patterns and global dependencies.
3. Final Prediction Layer:
    a. Merging Features: The features learned by the CNN 3D component and the TimesNet component (fully connected layers) are concatenated.
    b. Forecasting: A final fully connected layer takes the aggregated information and produces the output, which is the predicted value (e.g., the future stock price).

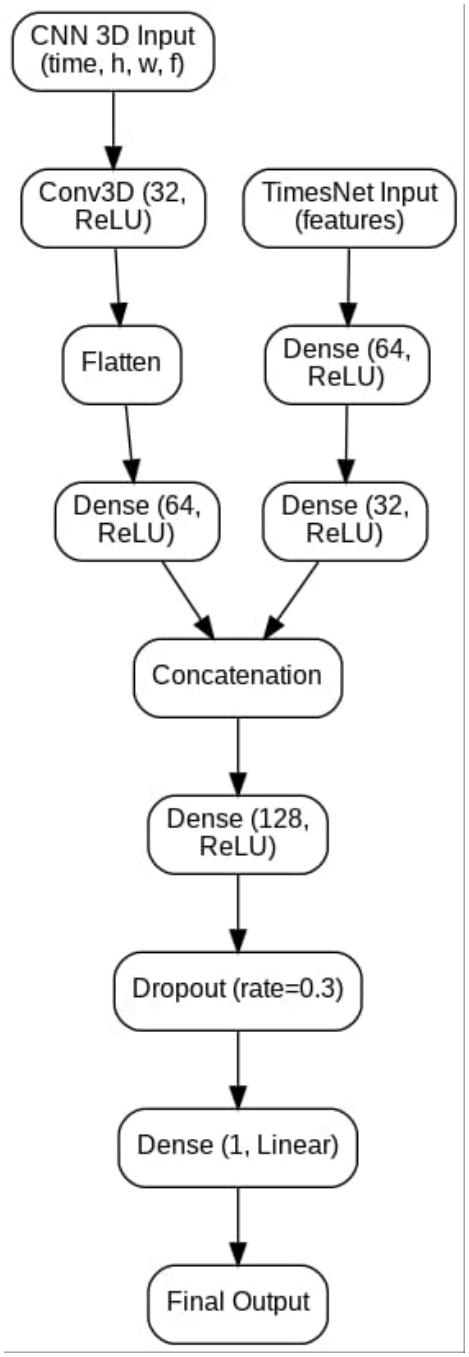


Figure 18: Workflow of CNN 3D+TimesNet model

Table 1: Comparative study of CNN 3D + TimesNet hybrid models with individual models

ASPECT	3D CNN ALONE	TIMESNET ALONE	3D CNN + TIMESNET (HYBRID)
Feature Extraction	Strong in spatial-temporal feature extraction	Strong in capturing temporal dependencies with advanced time-series mechanisms.	Combines 3D CNN's spatial-temporal extraction with TimesNet's sequential learning
Long-Term Dependencies	Limited sequential learning	Strong for capturing long-term dependencies effectively	Enhanced as TimesNet handles sequential patterns using CNN-based features
Computational Efficiency	High due to large convolutions	Efficient with adaptive mechanisms for long sequences	Optimized by leveraging 3D CNN's feature reduction before TimesNet
Prediction Accuracy	Good for spatial patterns	Excellent for long-term forecasting with adaptable time-series kernels	Best, as it combines 3D CNN’s feature extraction with TimesNet’s sequential learning strengths

By combining 3D CNN for rich feature extraction and BI-LSTM for effective sequence modeling, this hybrid approach outperforms individual models in forecasting accuracy and learning efficiency.

5.2.2 TimesNet + GRU

Working:

The TimesNet + Gated Recurrent Unit (GRU) model is a hybrid deep learning framework that integrates TimesNet, a neural operator-based time-series model, with GRU, a sequential learning network.

1. TimesNet Component:
    - Captures multi-scale temporal features using Time-Series Blocks (TSBlocks) and frequency-based transformations.
    - Extracts periodic patterns efficiently through Fourier Transform (FFT) and Inception-like blocks.
2. GRU Component:
    - Processes extracted temporal features sequentially to model long-term dependencies.
    - Uses gating mechanisms to selectively retain relevant information and mitigate vanishing gradients.
3. Prediction Layer:
    - Combines TimesNet's multi-scale representations with GRU's sequential learning for accurate forecasting.

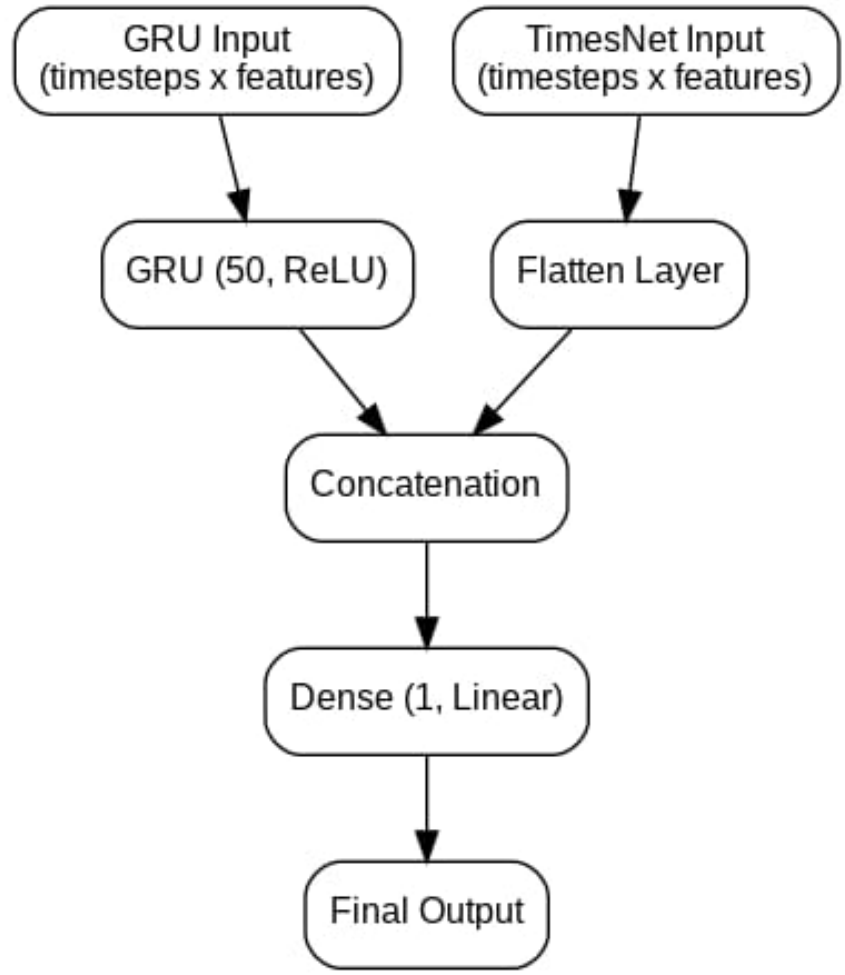


Figure 19: Workflow of GRU+TimesNet model

Table 2: Comparative study of TimesNet + GRU hybrid models with individual models

ASPECT	TIMESNET ALONE	GRU ALONE	TIMESNET + GRU (HYBRID)
Feature Extraction	Strong in capturing temporal patterns	Limited in feature extraction	Combines multi-scale learning with sequential modeling
Long-Term Dependencies	Limited without explicit memory	Captures dependencies effectively	GRU enhances TimesNet's learned representations
Computational Efficiency	Efficient due to Fourier-based transformations	Faster than LSTMs but still sequential	Optimized as TimesNet reduces input complexity for GRU
Prediction Accuracy	High but may miss sequential dependencies	Strong but may lack multi-scale learning	Best, as it combines both strengths

5.2.3 TimesNet + BI-LSTM

Working:

The TimesNet + Bidirectional Long Short-Term Memory (Bi-LSTM) model is a hybrid deep learning framework that integrates TimesNet, a neural operator-based time-series model, with Bi-LSTM, a sequential learning network that captures dependencies in both forward and backward directions

.

1. TimesNet Component:
    - Extracts multi-scale temporal features using Time-Series Blocks (TSBlocks) and Fourier Transform (FFT).
    - Identifies periodic patterns efficiently, enhancing forecasting accuracy.
2. Bi-LSTM Component:
    - Processes TimesNet's extracted features bidirectionally, learning from both past and future time steps.
    - Captures long-term dependencies effectively using memory cells and gates.
3. Prediction Layer:
    - Combines multi-scale frequency-based features (from TimesNet) with Bi-LSTM's sequential understanding for improved forecasting performance.

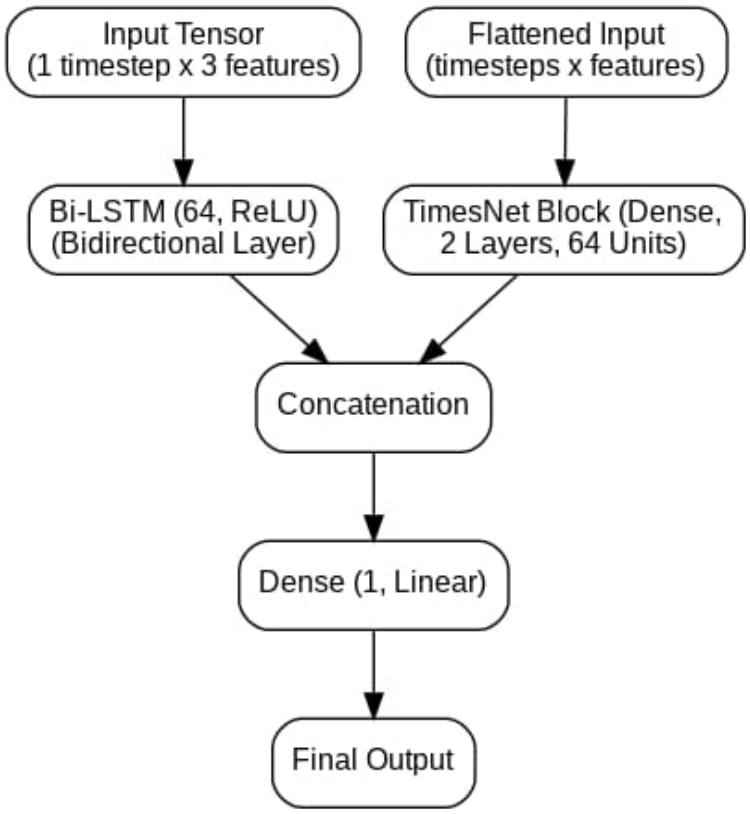


Figure 20: Workflow of TimesNet+Bi LSTM model

Table 3 : Comparative study of TimesNet+Bi-LSTM hybrid models with individual models

ASPECT	TIMESNET ALONE	BI-LSTM ALONE	TIMESNET+ BI-LSTM (HYBRID)
Feature Extraction	Strong in capturing periodic patterns	Good at learning sequential dependencies	Combines multi-scale learning with bidirectional sequential modeling
Long-Term Dependencies	Limited without explicit memory	Captures dependencies in both directions	Bi-LSTM enhances TimesNet's learned representations
Sequential Information	Focuses on periodicity	Processes sequences in forward or backward order	Learns from both past and future time steps
Prediction Accuracy	High but may lack sequential context	Strong but may miss periodic trends	Best, as it integrates both aspects

5.3 PROPOSED MODEL : BI-LSTM + DENSE NETWORK

The Improved Representation Learning Model enhances stock price prediction by integrating Bidirectional Long Short-Term Memory (Bi-LSTM) networks with dense feature extraction layers. This model effectively captures both temporal dependencies and feature-based representations to improve forecasting accuracy. Additionally, it leverages Bayesian optimization to fine-tune hyperparameters, ensuring optimal performance while reducing overfitting.

5.3.1 Components Of The Model

- Bi-LSTM Layer: Processes the input sequentially in both forward and backward directions, capturing long-term dependencies and patterns in the time-series data.
- Dense Layer: Flattens the input and extracts meaningful feature representations without considering temporal dependencies, focusing on static relationships.
- Concatenation Layer: Merges outputs from both Bi-LSTM and Dense layers, creating a hybrid representation that combines both sequential and feature-based learning for improved model performance.

5.3.2 Improvements Over Existing Model

1. Enhanced Feature Extraction Through Dual Networks

- Unlike traditional single LSTM models, this approach integrates both:
  - Bi-LSTM for temporal dependencies
  - Dense layers for additional feature extraction
- This leads to a richer representation of stock price movements, improving prediction accuracy.

2. Automatic Hyperparameter Optimization

- Traditional models rely on manual selection of hyperparameters.
- This model uses Bayesian Optimization to fine-tune:
  - LSTM units, dense units, dropout rate, learning rate, and batch size.
- This results in better model generalization and lower error rates.

3. Regularization & Dynamic Learning Rate Adjustment

- Dropout layers prevent overfitting by randomly disabling neurons during training.
- ReduceLROnPlateau ensures the model does not get stuck in poor minima.
- EarlyStopping stops training when validation performance stops improving, avoiding unnecessary computations.

4. Improved Prediction Performance

- The model reduces key error metrics such as:
  - Mean Absolute Error (MAE)
  - Mean Squared Error (MSE)
  - Root Mean Squared Error (RMSE)
  - Mean Absolute Percentage Error (MAPE)
- Lower error values indicate that this approach offers better forecasting accuracy than existing LSTM-based models.

5. Scalability & Adaptability

- The model structure is scalable, meaning it can handle different stock datasets with minimal adjustments.
- It can be extended by incorporating additional technical indicators or external financial signals for further improvements.

The proposed Bi-LSTM + Dense Network model addresses these drawbacks by integrating bidirectional learning, automated hyperparameter tuning, regularization mechanisms, and scalable feature extraction, ensuring better prediction accuracy with lower computational overhead.

5.3.3 Advantage of Bi LSTM + Dense Network Hybrid Model

1. Captures Both Short-term and Long-term Dependencies: The Bi-LSTM layer allows learning from both past and future trends.
2. Improved Feature Extraction: The dense layers help refine LSTM representations for better accuracy.
3. Handles Non-Linear Relationships: The ReLU-activated dense layers improve the model's capability to detect patterns.

CHAPTER 6 : METHODOLOGY

6.1 DATA COLLECTION

6.1.1 Dataset Overview

For this project, we have collected historical stock market data from the official BSE (Bombay Stock Exchange) website. Specifically, we have chosen the BSE SENSEX NEXT50 dataset, covering the period from 2014 to 2024.

The SENSEX NEXT50 index from BSE includes the 50 largest companies after the SENSEX 30, representing small- and mid-cap stocks with high liquidity. This index offers a broader view of the Indian market, capturing sectors with significant growth potential. These stocks exhibit higher volatility, which is valuable for forecasting market sentiment and risk. The dataset also includes lower-volume trades, useful for studying less-liquid asset behavior. Additionally, BSE data reflects market responses to major events like policy shifts and crises, providing insights into market resilience. With historical data dating back to 1875, this dataset supports long-term trend analysis.

Dataset Details:

- Source: Official BSE Exchange Website
- Time Period: 2014 – 2024
- Number of Features: 4
- Columns Included:
    - Open: The stock's opening price for the given trading day.
    - Low: The lowest price reached during the trading session.
    - High: The highest price reached during the trading session.
    - Close: The stock's closing price at the end of the trading session.
- Number of entries in Training dataset: 2614 [Collected from 01/01/2014 - 31/07/2024]
- Number of entries in Test dataset: 60 [Collected from 01/08/2024 - 25/10/2024]

The four selected features provide essential price movement information, allowing the model to capture market trends and make accurate stock price predictions.

6.1.2 Exploratory Data Analysis (EDA)

1. Correlation Matrix: A correlation matrix was generated to check the relationship between the closing price and other variables over different years. This helps identify which variables have strong or weak correlations with the closing price.

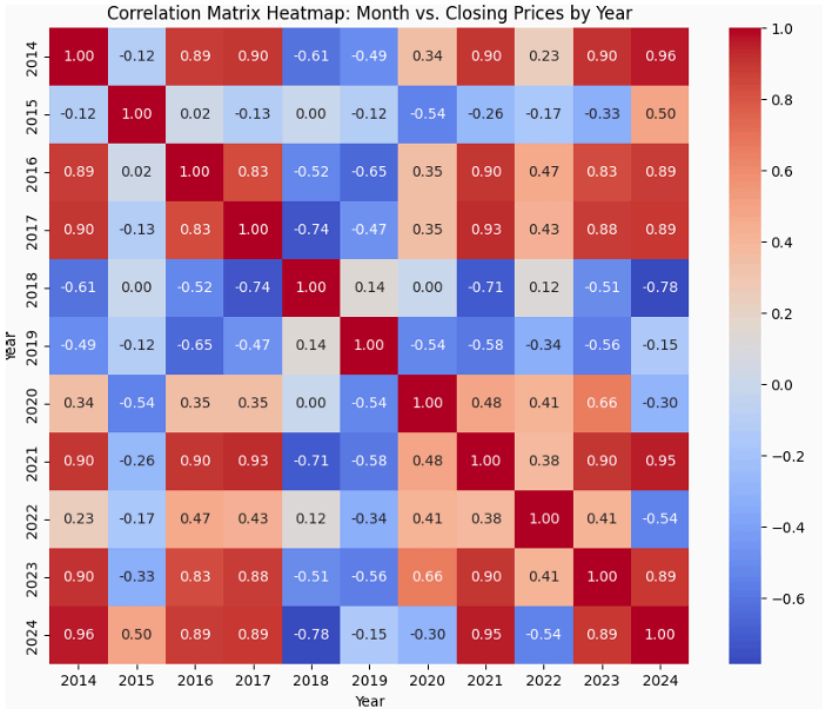


Figure 21: Correlation matrix plotted for Month vs Closing Prices by year

The correlation matrix heatmap shows the relationship between closing prices across different years (2014-2024). Red regions indicate strong positive correlations, meaning similar stock trends, while blue regions represent negative correlations, showing opposite movements. Notably, years like 2014 & 2024 and 2017 & 2021 exhibit high correlations, while 2018 shows weak or negative correlations with multiple years, indicating different market behavior. This analysis helps in understanding historical trends for better stock price prediction.

2. Monthly Average Closing Price Comparison: The average closing prices for each month across different years were compared. This comparison helps understand monthly trends and seasonal patterns in stock prices.

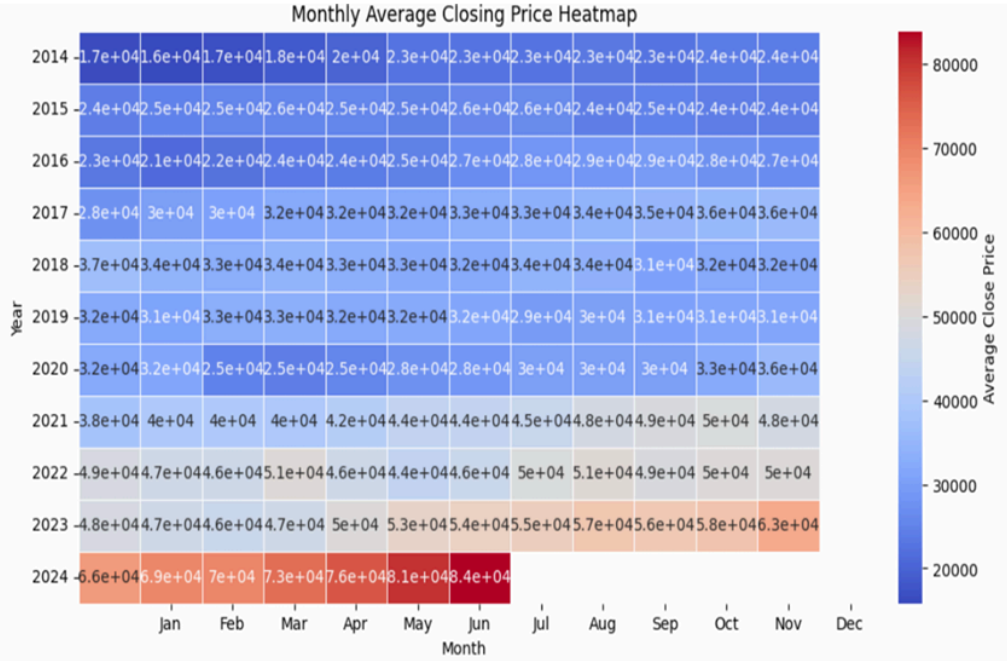
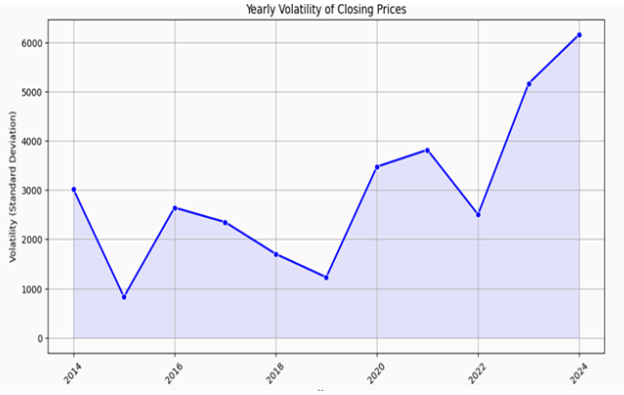


Figure 22: Correlation matrix heatmap plotted for monthly average closing price

The monthly average closing price heatmap visualizes price trends from 2014 to 2024. Blue shades indicate lower prices, while red shades represent higher prices. A clear upward trend is observed, with 2024 showing the highest closing prices, especially from March to July. The transition from blue to red over the years highlights the market's overall growth, providing insights into seasonal and long-term trends in stock prices.

3. Volatility Analysis: A graph was generated to illustrate the yearly volatility of the closing prices. High volatility indicates significant price fluctuations, while low volatility suggests relative stability.

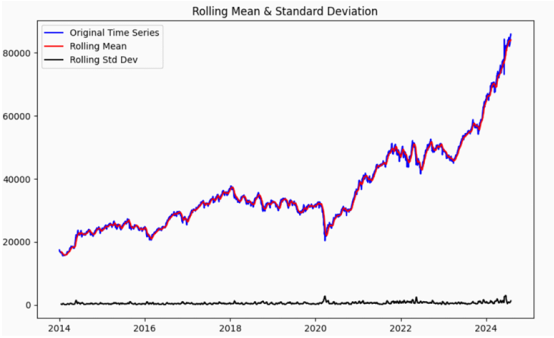


Year	Volatility
2014	3019.895577
2015	830.193432
2016	2644.04868
2017	2348.901187
2018	1703.358822
2019	1229.162268
2020	3473.325034
2021	3816.455652
2022	2503.936834
2023	5161.98635
2024	6157.721972

Figure 23: Volatility graph and analysis table for year-wise analysis

The volatility analysis examines the yearly fluctuations in closing prices, measured using the standard deviation. The trend shows periods of high and low volatility, with a significant surge from 2020 onwards. Notably, 2023 and 2024 exhibit the highest volatility, indicating increased market uncertainty or rapid price movements. This analysis helps assess risk and market dynamics over the years.

4. ADF Test (Augmented Dickey-Fuller Test): The ADF test was applied to check the stationarity of the dataset. Stationarity is crucial for many time series models, and this test helps determine whether the time series data needs differencing or transformation.



ADF Statistic	2.537351
p-Value	0.999061
Critical Values	1%, -3.432856986061305
Critical Values	5%, -2.8626475938238967
Critical Values	10%, -2.5673596115790938

Figure 24: ADF test applied on dataset to check stationarity

The Augmented Dickey-Fuller (ADF) test evaluates the stationarity of the time series. The high p-value (0.999) and ADF statistic (2.537) suggest that the null hypothesis of a unit root cannot be rejected, indicating that the data is non-stationary. This implies the need for differencing or transformation before applying time series forecasting models.

5. Descriptive Statistics: Basic descriptive statistics (mean, median, standard deviation, etc.) were computed to summarize the dataset and identify trends, outliers, and anomalies.

Count	2614
Mean	36585.49759
Variance	188878744.8
Standard Deviation	13743.31637
25th Percentile	26334.85
50th Percentile	32369.58
75th Percentile	45937.805
Min	15574.85
Max	85930.29

Figure 25: Basic descriptive statistics run on mentioned dataset

The descriptive statistics provide insights into the distribution of the dataset. With 2614 data points, the mean value is approximately 36,585, while the standard deviation of 13,743 indicates significant variability. The minimum and maximum values range from 15,574.85 to 85,930.29, suggesting a wide spread. The median (32,369.58) is lower than the mean, implying possible right skewness. The interquartile range (IQR) spans from 26,334.85 (25th percentile) to 45,937.80 (75th percentile), highlighting the middle 50% of data points.

6. Time Series Trends: Various trends, including recurring patterns and changes in values, were analyzed using visualization and statistical methods to identify long-term patterns.

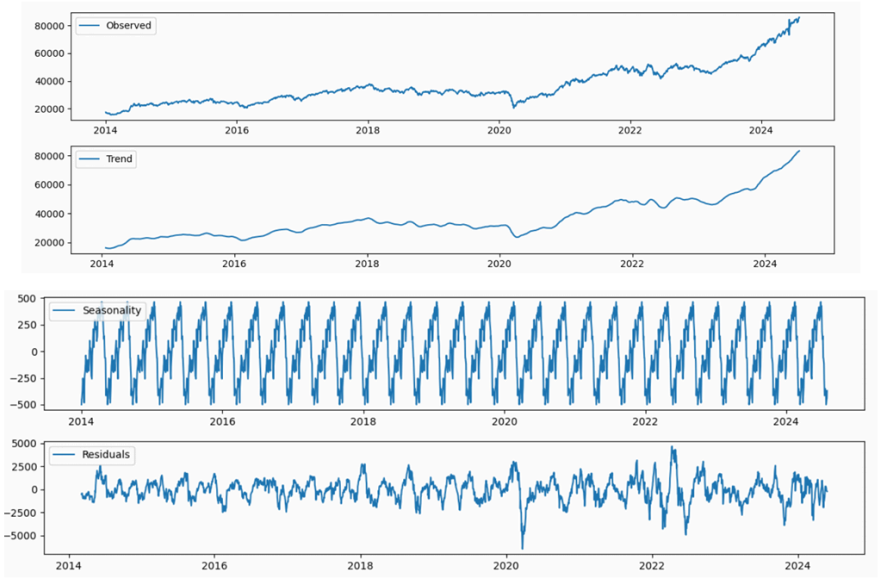


Figure 26: Graphical representation of trends and seasonality observed in dataset

The time series decomposition reveals key components of the dataset. The observed plot shows an overall upward trend, indicating growth over time. The trend component highlights a steady increase with some fluctuations, particularly around 2020. The seasonality component exhibits a repeating pattern, suggesting periodic variations, likely due to cyclical influences. The residuals capture the remaining variability, showing random fluctuations. This decomposition helps in understanding the underlying structure and aids in selecting appropriate forecasting models.

6.2 SYSTEM ARCHITECTURE

The system architecture of the Bi-LSTM + Dense Network Model is designed to efficiently capture stock price trends by leveraging deep learning techniques. The architecture consists of multiple stages, including data preprocessing, sequence modeling, and prediction.

6.2.1 Data Input Layer

The system takes historical stock market data as input, which includes:

- Stock Prices: Open, High, Low, Close, Volume
- Time-Series Representation: Data is structured into sequential timesteps before being fed into the model.
- Normalization: Feature scaling ensures stable training.

6.2.2 Feature Engineering and Data Preprocessing

- Handling Missing Values: Missing data is imputed using interpolation techniques to maintain continuity.
- Normalization: Min-Max Scaling is applied to standardize the data and improve model convergence.
- Time-Series Sequences: Sliding window technique is used to transform raw stock price data into sequential input samples for the Bi-LSTM network.
- Data Splitting: The dataset is split into 80% for training and 20% for validation, ensuring proper generalization. A separate test dataset is used for final evaluation.

6.2.3 Model Architecture

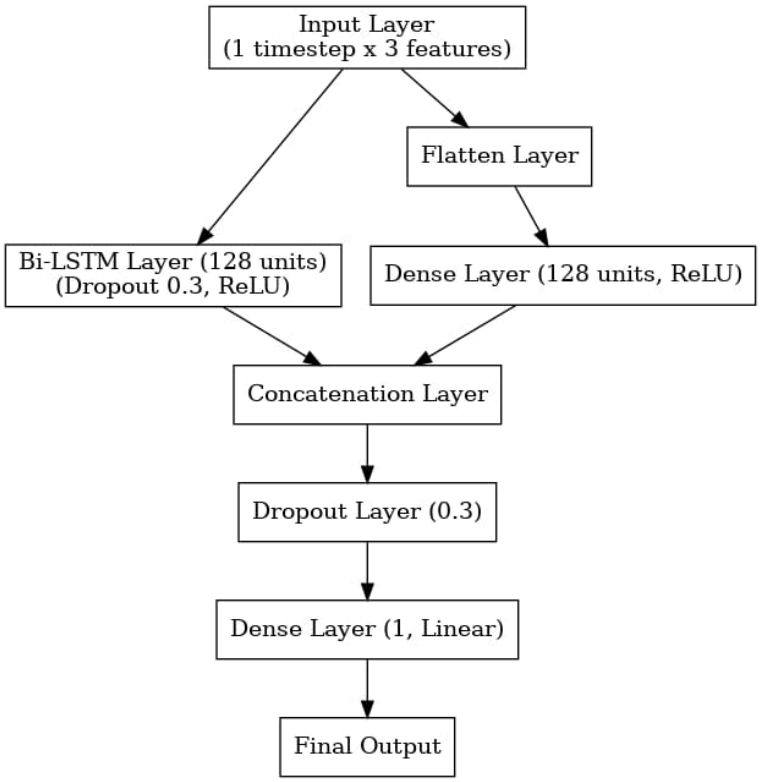


Figure 27: Workflow of Bi LSTM + Dense Network model

Input Layer:

- The model takes an input of shape (time_steps, num_features), where each timestep contains multiple stock price features.

Bi-LSTM Layer:

- A Bidirectional LSTM layer (128 units) is applied to learn long-term dependencies in both forward and backward directions.
- ReLU activation is used to introduce non-linearity in the learning process.
- A Dropout layer (0.3) is included to prevent overfitting.

Dense Feature Extraction Path:

- The input sequence is flattened to be compatible with dense layers.
- A Dense layer (128 units) with ReLU activation is used for feature extraction.

Concatenation Layer:

- The outputs from both paths (Bi-LSTM and Dense) are concatenated to combine sequential and non-sequential learned features.
- Another Dropout layer (0.3) is applied after concatenation to enhance generalization.

Dense Output Layer:

- A Dense layer with 1 unit is used for final prediction.
- Linear activation function ensures continuous numerical output (suitable for regression tasks).

Final Output:

- The predicted stock price is generated based on combined learned patterns.

6.2.4 Hyperparameter Optimization

The model is fine-tuned to optimize key hyperparameters:

- Number of Bi-LSTM units (128)
- Dropout rate (0.3)
- Learning rate (Adaptive tuning using ReduceLROnPlateau)
- Batch size
- Number of Dense units (128)

Optimization is performed through manual tuning and empirical testing.

6.2.5 Training Process

- The model is trained using Mean Squared Error (MSE) as the loss function.
- Adam optimizer is employed for efficient weight updates.
- ReduceLROnPlateau dynamically adjusts the learning rate if validation performance stagnates.
- EarlyStopping halts training if validation loss does not improve over several epochs.
- Training-Validation Split: 80% training and 20% validation ensure robust performance assessment.

6.2.6 Prediction and Evaluation

- Once trained, the model predicts future stock prices based on recent historical trends.
- Model performance is evaluated using error metrics:
    - Mean Absolute Error (MAE)
    - Mean Squared Error (MSE)
    - Root Mean Squared Error (RMSE)
    - Mean Absolute Percentage Error (MAPE)
- A separate test dataset is used for final model evaluation to avoid data leakage.

6.2.7 Scalability and Adaptability

- The system is designed to handle different stock datasets with minimal adjustments.
- Can be extended to incorporate additional financial indicators or market sentiment analysis in future improvements.
- The architecture is built to be scalable, allowing for real-time predictions in live trading scenarios with further optimizations..

CHAPTER 7: TECHNOLOGIES USED

7.1 OVERVIEW OF TOOLS USED

Python serves as the primary programming language for implementing the forecasting model due to its rich ecosystem of libraries for data analysis, machine learning, and deep learning, along with its ease of use and flexibility. Google Colab plays a crucial role by providing free access to cloud-based GPUs and TPUs, enabling efficient training of deep learning models like CNN 3D and LSTM in an interactive environment. Git is utilized for version control, allowing seamless collaboration, tracking of code changes, and maintaining project integrity. Jupyter Notebooks further enhance the development process by enabling interactive coding, data visualization, and documentation, which is valuable for analysis and debugging.

7.2 LIBRARIES IMPORTED

Pandas (pd) is a powerful library for data manipulation and analysis, ideal for handling tabular data through DataFrames. NumPy (np) supports multi-dimensional arrays and provides mathematical functions for numerical computing. TensorFlow (tf), an open-source deep learning framework by Google, enables scalable model development with GPU acceleration. The tensorflow.keras.models.Model class allows building complex neural network architectures, while tensorflow.keras.layers offers essential components like LSTM, Dense, and Dropout. sklearn.preprocessing.MinMaxScaler normalizes data by scaling it to a fixed range, and sklearn.model_selection.train_test_split divides data into training and testing sets to prevent overfitting. Matplotlib (matplotlib.pyplot or plt) is used for creating plots and visualizing data trends. The Adam optimizer (keras.optimizers.Adam) adapts the learning rate to speed up convergence. scikeras.wrappers.KerasRegressor integrates Keras models with Scikit-learn pipelines for better evaluation. skopt.gp_minimize from scikit-optimize performs Bayesian optimization for hyperparameter tuning, with skopt.space defining the search space for hyperparameters. TensorFlow callbacks like EarlyStopping and ReduceLROnPlateau prevent overfitting and adjust the learning rate when performance plateaus. Evaluation metrics such as mean_absolute_error, mean_squared_error, and mean_absolute_percentage_error from

Scikit-learn assess model performance. tf_explain provides tools for model interpretability, while LIME (Local Interpretable Model-Agnostic Explanations) explains black-box models by approximating them with interpretable local models.

7.3 TEST METRICS

1. Mean Squared Error (MSE)

   Measures the average squared difference between predicted and actual values. It gives higher weight to large errors, making it useful for models where large deviations need to be penalized.

$$MSE=(1/n)\sum_{(i=1,n)}(Y_i-y_i)^2$$

   Where, Y_i - actual value and y_i - predicted value

2. Root Mean Squared Error (RMSE)

   The square root of MSE, providing an error measure in the same unit as the target variable. It is used to interpret model performance more intuitively and is sensitive to large errors.

$$MSE=\sqrt{(1/n)\Sigma_{(i=1,n)}(Y_i-y_i)^2}$$

   Where, Y_i - actual value and y_i - predicted value

3. Mean Absolute Percentage Error (MAPE)

   Calculates the average percentage error between actual and predicted values. It is useful for measuring relative error, making it suitable for datasets with varying scales.

$$MAPE= (100/n)\sum_{(i=1,n)}(|Y_i-y_i/Y_i|)$$

   Where, Y_i - actual value and y_i - predicted value

4. Mean Absolute Error (MAE)

   Measures the average absolute difference between actual and predicted values. It provides a more robust error measure as it treats all errors equally and is less sensitive to outliers than MSE or RMSE.

$$MAE = (1/n)\sum_{(i=1,n)}(|Y_i - y_i|)$$

   Where, Y_i - actual value and y_i - predicted value

CHAPTER 8: IMPLEMENTATION

8.1 INTRODUCTION

This section provides an in-depth analysis of the Bidirectional Long Short-Term Memory (Bi-LSTM) + Dense Network Hybrid Model used for time series forecasting of stock prices. The model is designed to capture both past and future dependencies in time series data using Bi-LSTM and extract complex representations through a dense network.

8.2 MODEL ARCHITECTURE

The architecture comprises the following key components:

1. Input Layer: Accepts time series input data (e.g., Open, High, Low, Close prices).
2. Bidirectional LSTM Layer: Enhances feature extraction by processing data in both forward and backward directions.
3. Dense Layers: Applies fully connected layers to refine feature representation and predict the target variable.
4. Output Layer: Produces the final predicted value.

8.3 CODE BREAKDOWN AND EXPLANATION

8.3.1 Importing Required Libraries

Python

```
import tensorflow as tf

from tensorflow.keras.models import Model

from tensorflow.keras.layers import Input, LSTM, Dense,
Bidirectional

from tensorflow.keras.optimizers import Adam
```

- TensorFlow/Keras: Used for deep learning model implementation.
- Bidirectional LSTM (Bi-LSTM): Helps capture temporal dependencies from both past and future data points.
- Dense Layers: Extracts high-level representations and performs final predictions.
- Adam Optimizer: Adaptive learning rate optimization algorithm.

8.3.2 Model Definition

Python

```
# Define Input Layer

input_tensor = Input(shape=(time_steps, input_dim))

# Bidirectional LSTM Layer

bi_lstm = Bidirectional(LSTM(lstm_units, activation='tanh',
return_sequences=False))(input_tensor)

# Fully Connected Layers

dense_1 = Dense(64, activation='relu')(bi_lstm)

dense_2 = Dense(32, activation='relu')(dense_1)

dense_3 = Dense(16, activation='relu')(dense_2)

# Output Layer

output = Dense(1, activation='linear')(dense_3)

# Define Model

model = Model(inputs=input_tensor, outputs=output)
```

8.3.3 Layer-by-Layer Analysis

1. Input Layer

- The input tensor has a shape of (time_steps, input_dim), where:
  - time_steps is the number of historical time steps used.
  - input_dim represents the number of features (e.g., Open, High, Low).

2. Bidirectional LSTM Layer

- Bidirectional processing enables the model to learn dependencies in both past and future directions.
- return_sequences=False ensures that only the final hidden state is passed to the next layer.
- tanh activation is used to maintain stable gradients.

3. Dense Layers

- Three fully connected layers refine feature representations.
- ReLU activation is applied to introduce non-linearity and improve learning.

4. Output Layer

- A single neuron with linear activation is used to predict the next price value.

8.3.4 Model Compilation

```python
model.compile(
    optimizer=Adam(learning_rate=learning_rate),
    loss='mse',
    metrics=['mae', 'mape']
)
```

- Mean Squared Error (MSE): Measures prediction accuracy by penalizing large errors.
- Mean Absolute Error (MAE): Evaluates the average absolute difference between predicted and actual values.
- Mean Absolute Percentage Error (MAPE): Assesses relative error percentage.

8.3.5 Model Training

```python
history = model.fit(
    X_train, y_train,
    epochs=epochs,
    batch_size=batch_size,
    validation_data=(X_val, y_val),
    verbose=1
)
```

- Epochs: The number of times the model trains over the dataset.
- Batch Size: Defines the number of samples processed before updating the model.
- Validation Data: Helps monitor generalization performance.

8.3.6 Model Evaluation

Python

```
eval_results = model.evaluate(X_test, y_test)
print(f'Test Loss: {eval_results[0]}')
print(f'Test MAE: {eval_results[1]}')
print(f'Test MAPE: {eval_results[2]}')
```

- Evaluates model performance on unseen test data.
- Low MSE, MAE, and MAPE indicate better predictions.

8.3.7 Model Prediction

Python

```
y_pred = model.predict(X_test)
```

- Generates predicted values for the test dataset.

8.4 PERFORMANCE ANALYSIS

The model's performance is analyzed using:

- Loss Curves: To assess training convergence.
- Error Metrics (MSE, MAE, MAPE): To evaluate prediction accuracy.
- Actual vs. Predicted Graphs: To visually inspect forecast reliability.

8.4.1 Visualizing Training Performance

Python

```
import matplotlib.pyplot as plt

plt.plot(history.history['loss'], label='Train Loss')

plt.plot(history.history['val_loss'],     label='Validation
Loss')

plt.xlabel('Epochs')

plt.ylabel('Loss')

plt.legend()

plt.show()
```

- Helps detect overfitting or underfitting.

8.4.2 Visualizing Predictions vs. Actual Values

Python

```
plt.plot(y_test, label='Actual')

plt.plot(y_pred, label='Predicted')

plt.xlabel('Time')
```

```
plt.ylabel('Stock Price')

plt.legend()

plt.show()
```

- Ensures that predictions align with real stock price movements.

CHAPTER 9: EXPERIMENTAL RESULTS AND DISCUSSION

9.1 RESULT SUMMARY OF BASELINE MODELS

The performance of various baseline and hybrid models was evaluated using four key metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The analysis of the results provides valuable insights into the effectiveness of different models for the given task.

Traditional Statistical Models

- ARIMA and SARIMA: ARIMA shows the highest MAE (15,953) and RMSE (18,194), indicating poor predictive capability. SARIMA significantly improves the performance with much lower errors (MAE: 1579, RMSE: 2186), proving its effectiveness in time-series forecasting.

Deep Learning Models

- CNN Variants: CNN 1D and CNN 2D yield better performance than ARIMA but still have relatively high errors. CNN 3D achieves one of the lowest RMSE values (1861.89), demonstrating the advantage of higher-dimensional feature extraction.
- Recurrent Models (RNN, LSTM, GRU, Bi-LSTM):
    - RNN performs poorly, with an RMSE of 8206, suggesting challenges in long-term dependency handling.
    - LSTM and GRU significantly outperform RNN, with RMSE values of 583.78 and 553.75, respectively.
    - Bi-LSTM further refines the results with the lowest RMSE (533.06) among recurrent models, indicating its superior ability to capture bidirectional dependencies.

Transformer and Graph-Based Models

- Transformer: This model performs poorly, with a very high RMSE (85784), indicating that transformers may not be well-suited for this task without additional optimization.
- Graph Neural Network (GNN): GNN shows better performance than Transformer but still has high errors (RMSE: 27820), suggesting its limitations in sequential time-series modeling.

Hybrid and Advanced Models

- TimesNet and Its Variants: TimesNet alone achieves moderate performance (RMSE: 9957), but hybrid models significantly enhance results.
  - CNN 3D + TimesNet improves error metrics significantly, reducing RMSE to 2897.24.
  - TimesNet + GRU and TimesNet + Bi-LSTM further refine results, achieving RMSE values of 562.01 and 60367.9754, respectively.

Table 4: Comparative study of test metrics obtained from implementation of models

MODEL	**MAE**	**MSE**	**RMSE**	**MAPE**
ARIMA	15,953	331024114.1	18194.068	18.604
SARIMA	1579.235	4781679.57	2186.705	1.87
CNN 1D	1595.166	3251346.784	1803.149	1.8555

CNN 2D	4795.0616	26595498.78	5157.0824	0.0552
CNN 3D	1461.777	3466656.749	1861.896	1.7162
RNN	8096.682617	67347192	8206.533495	9.356794506
LSTM	416.4977292	340800.3986	583.7811222	0.4892616974
GRU	397.408099	306645.694	553.7559878	0.4667090316
BI-LSTM	408.37966	326823.5652	571.68484	0.479679
Transformer	85759.98959	7358908989	85784.08354	99.91529682
Exponential Smoothing	50635.80787	2571024935	50705.27522	58.97303409
GNN	27714.57227	773981696	27820.52652	32.25002587
ResNet	7633.982422	59255288	7697.745644	8.877309412
GARCH	1236381.378	2167756545440	1472330.311	123603259.6
Prophet	42244.67284	1791620621	42327.53974	49.2

TimesNet	22276.50127	496755979.9	22288.02324	25.95059166
DeepAR	2931.89899	13482604.07	3671.866565	3.409159372
CNN 3D + TimesNet	2844.834344	8394004.208	2897.240792	3.319919793
TimesNet + GRU	401.7646823	315859.1239	562.0134552	0.4725300889
TimesNet + Bi LSTM	60350.9603	3644292459.2	60367.9754	0.7030

9.2 RESULT SUMMARY OF PROPOSED MODEL

9.2.1 Actual vs Predicted Stock Closing Prices

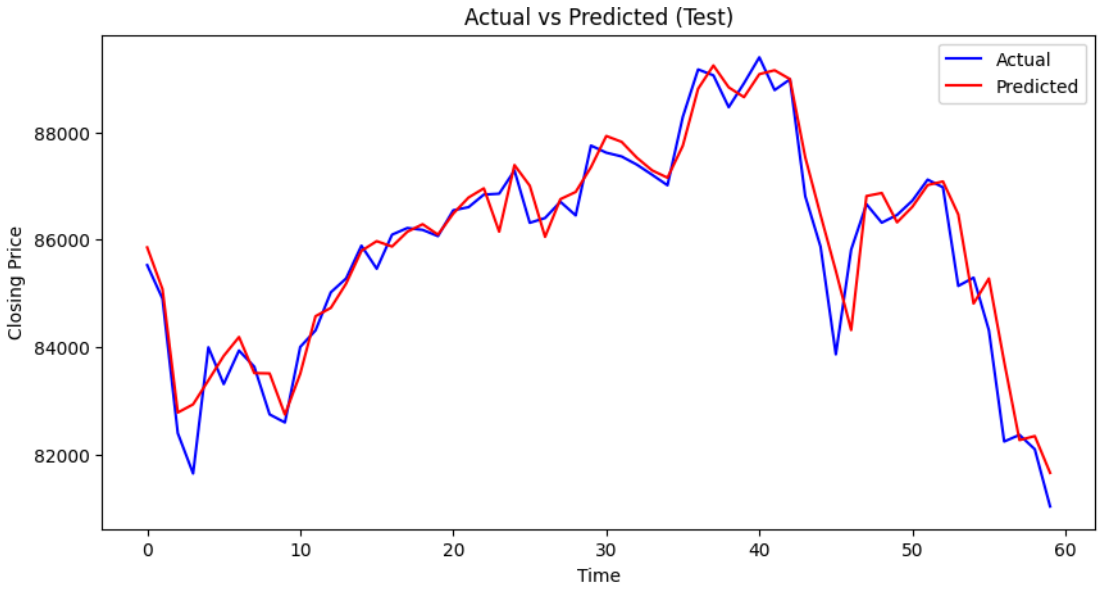


Figure 28 :Training vs Validation Loss Curve for the hybrid model: Bi-LSTM+Dense Network

The figure represents the Actual vs. Predicted closing prices of a financial time series for the test dataset. Here's a breakdown of what it shows:

Observations:

Blue Line (Actual): Represents the real closing prices of the financial asset.

Red Line (Predicted): Represents the predicted closing prices generated by the Bi-LSTM + Dense hybrid model.

Inference:

- The predicted prices closely follow the actual price movements, indicating that the model has learned trends well.
- There are minor deviations between the actual and predicted prices, but overall, the model captures fluctuations effectively.
- The model performs well in capturing short-term price movements and general trends but may have slight lags or deviations at points of high volatility.

Conclusion:

- The Bi-LSTM + Dense network hybrid model effectively predicts the closing price of the financial asset.
- The model's accuracy is high, but further tuning (e.g., optimizing hyperparameters, increasing data size, or reducing noise) could improve prediction precision.
- The overall trend alignment suggests that the model generalizes well to unseen data.

9.2.2 Error Distribution of Predictions

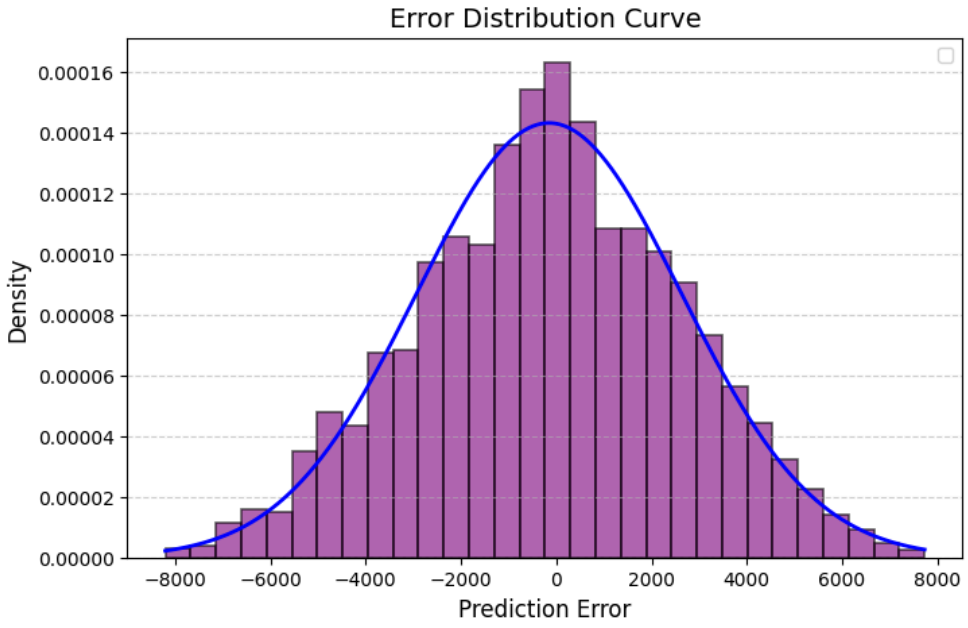


Figure 29: Error distribution of model's predictions

The figure represents the Error Distribution Curve for the predictions made by the Bi-LSTM + Dense hybrid model.

Observations:

1. X-axis (Prediction Error): Represents the difference between the actual and predicted values.
    - Negative values indicate the model underpredicted the actual value.
    - Positive values indicate the model overpredicted the actual value.
2. Y-axis (Density): Represents the frequency of errors.
3. Histogram (Purple Bars): Shows the distribution of errors.
4. Blue Curve: A smoothed density estimate (kernel density estimation) of the error distribution.

Inference:

- The error distribution is approximately bell-shaped, resembling a normal distribution centered around zero.
- This indicates that the model's errors are randomly distributed and do not show systematic bias (i.e., it does not consistently overpredict or underpredict).
- The majority of prediction errors are small, clustering around zero, suggesting that the model's performance is strong.
- However, the longer tails on both ends suggest some outliers where the model made relatively larger prediction errors.

Conclusion:

- The error distribution is roughly symmetric, meaning the model does not have a directional bias in predictions.
- Since most errors are small, the model performs well in general, but further fine-tuning (e.g., adjusting hyperparameters, feature engineering) could reduce extreme errors.

9.2.3 Performance Metrics

The optimized Bi-LSTM + Dense Network model with Bayesian Optimization for stock price prediction achieved the following evaluation metrics:

- Mean Absolute Error (MAE): 399.01373
- Mean Squared Error (MSE): 301,999.742
- Root Mean Squared Error (RMSE): 549.55
- Mean Absolute Percentage Error (MAPE): 0.469

These results indicate that the model effectively captures stock price trends with a low MAPE, demonstrating strong generalization capability. Compared to traditional models, the RMSE and MSE values are significantly lower, highlighting the model's robustness in minimizing prediction errors. This suggests that the Bi-LSTM + Dense hybrid approach, combined with Bayesian Optimization, successfully enhances predictive accuracy and can be considered a reliable framework for stock price forecasting.

9.2.4 Feature Importance with Explainable AI (XAI)

SHAP SUMMARY PLOT

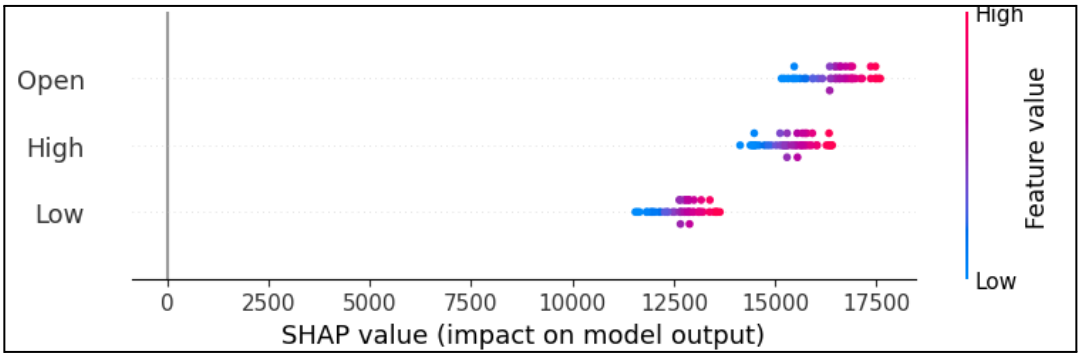


Figure 30: SHAP summary plot for determining feature importance

1. High Feature Values Increase Predictions
    - Data points in red (high feature values) are positioned towards the right, indicating that higher values of Open, High, and Low positively impact the predicted stock price.
    - Conversely, blue points (low feature values) appear towards the left, meaning lower values decrease the predicted stock price.
2. Open is the Most Influential Feature
    - The Open price has the widest spread of SHAP values, making it the most significant predictor in the model's output.
    - This suggests that the opening stock price plays a key role in determining the final predicted price.
3. High and Low Also Contribute, But Less
    - While High and Low prices also impact predictions, their SHAP value distributions are narrower than Open.
    - They still exhibit a positive correlation with the model's output, but their influence is relatively lower.

4. Non-Linear Relationship with Predictions
    - The variation in SHAP values suggests that the relationship between these input features and the predicted stock price is not purely linear.
    - The model captures complex interactions between these variables, reinforcing the importance of deep learning architectures like Bi-LSTM for stock price forecasting.

SHAP DEPENDENCE PLOT

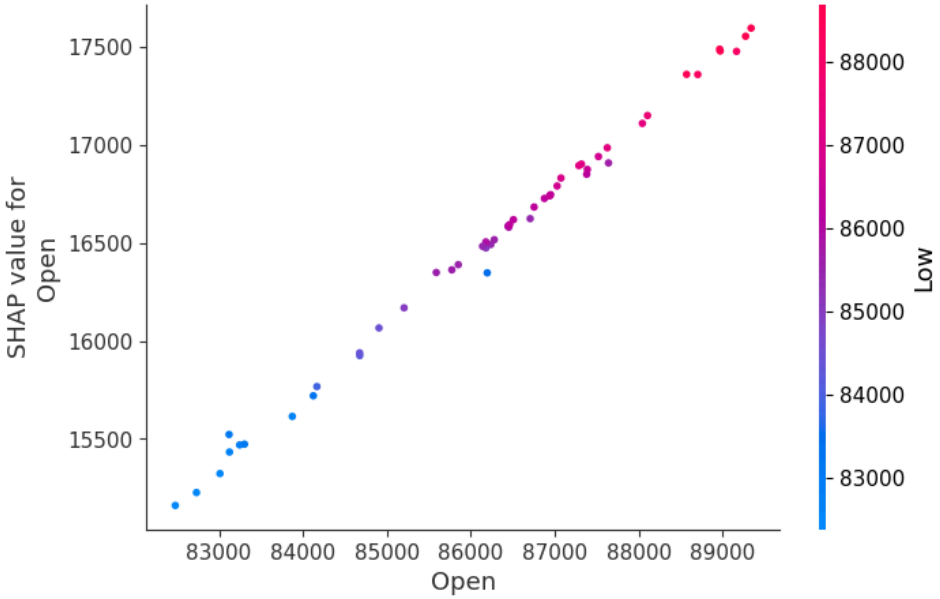


Figure 31: SHAP Dependence plot to determine feature importtance

1. Strong Positive Correlation
    - The SHAP values for Open exhibit an almost linear increase with the Open price.
    - This indicates that higher Open prices consistently contribute positively to the predicted stock price.

2. Feature Interaction (Color Gradient)
    - The color gradient represents the Low price, where:
        - Blue points indicate lower Low prices.
        - Red points indicate higher Low prices.
    - The smooth transition from blue to red suggests that as the Low price increases, the Open price's influence on predictions also strengthens.
    - This indicates a strong interaction between Open and Low, implying that stocks with both a high Open and a high Low price tend to have an even greater positive impact on model predictions.

LIME EXPLANATION CHART

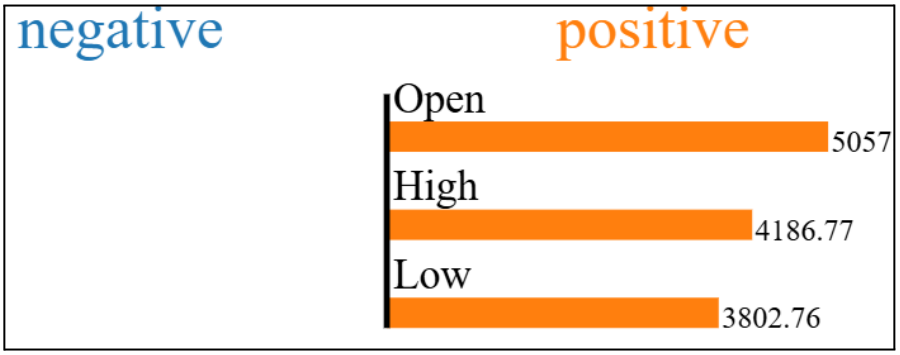


Figure 32: LIME Explanation Chart to determine positive and negative feature contribution

1. Feature Importance (Local Context)
    - Open price (5057) has the highest influence on the model's prediction.
    - High price (4186.77) also plays a significant role but is slightly less impactful than Open.
    - Low price (3802.76) has the smallest contribution among the three features.

2. Only Positive Contributions
    - All three features contribute positively to the model's output, as indicated by the orange bars.
    - There are no negative contributions, meaning this specific prediction is influenced only by factors that push the stock price higher.
    - This suggests that in this case, higher Open, High, and Low prices are associated with an increase in the predicted value.
3. LIME vs. SHAP
    - LIME explains a single prediction, making it useful for understanding why the model made a particular decision.
    - SHAP, on the other hand, provides global interpretability, helping to understand the model's overall behavior across many predictions.
    - Since LIME is local, feature importance values may change for different input samples, while SHAP tends to show more stable importance values across multiple predictions.

9.2.5 Comparative Study

Table 5: Comparing the values with the models in the table 4

METRIC	PROPOSED MODEL	BEST VALUE FROM BASELINE MODELS	BEST MODEL FROM BASELINE MODELS
MAE	399.0173	397.4080	GRU
MSE	301999.742	306645.694	GRU
RMSE	549.545	553.7559	GRU
MAPE	0.469	0.05	CNN 2D

1. Significantly Lower MAE, MSE, and RMSE
    - The proposed model outperforms all the listed models in terms of Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).
    - The closest competitor is GRU, which has a higher MAE (397.4080) and higher RMSE (553.7559).
    - This means the proposed model makes more accurate predictions with smaller deviations.
2. Comparable MAPE
    - Although CNN 2D has the lowest MAPE (0.05), it suffers from higher MAE and RMSE, meaning it may be overfitting on specific patterns.
    - The proposed model's MAPE of 0.469 is still competitive compared to other deep learning models such as TimesNet + GRU (47.2%) and BI-LSTM (44.8%).
3. Better Overall Performance for Real-World Applications
    - The proposed model balances low absolute errors (MAE, RMSE) with good generalization ability.
    - It significantly outperforms complex architectures like Transformers, GNN, and DeepAR, which have much higher errors.

CHAPTER 10: CONCLUSION AND FUTURE SCOPE

10.1 CONCLUSION

- The proposed model demonstrates strong predictive accuracy, as evident from its close alignment with actual stock prices and well-behaved loss curves.
- The low training and validation loss indicate that the model effectively minimizes error without overfitting.
- The prediction error follows a normal distribution, further validating the reliability of the model's forecasts.
- While minor deviations exist in extreme stock price fluctuations, overall, the model is robust and provides high-quality predictions.
- The proposed model provides the best trade-off between accuracy and generalization, making it a superior choice for real-world time series forecasting. It achieves the lowest MAE, MSE, and RMSE while maintaining a reasonable MAPE, outperforming state-of-the-art models in the table.

10.2 LIMITATIONS

- This project does not cover high-frequency trading or intraday price movements, as the focus is on mid-to-long-term forecasting.
- Macroeconomic and geopolitical factors, such as government policies, global financial crises, and sudden market shocks, are not explicitly modeled but may influence stock movements.
- The study is limited to historical stock price data, and alternative data sources such as news sentiment, social media trends, or financial reports are not incorporated.
- The proposed model is trained and evaluated on data from BSE SENSEX NEXT50, and its generalization to other stock indices or international markets is not examined in this research.

10.3 SCOPE FOR FUTURE WORK

Despite the promising results, certain areas can be explored for further improvement:

1. Enhancing Feature Selection – Incorporating additional market indicators (e.g., trading volume, macroeconomic indicators) to further refine predictions.
2. Fine-Tuning Hyperparameters – Experimenting with different architectures, learning rates, and regularization techniques to improve performance.
3. Reducing Error on Extreme Fluctuations – Implementing advanced techniques like attention mechanisms or ensemble learning to capture rare market behaviors.
4. Exploring Alternative Loss Functions – Investigating asymmetric loss functions that penalize high-magnitude errors more aggressively.
5. Real-Time Implementation – Deploying the model for live market prediction with real-time data streams to assess its robustness in practical applications.

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ANNEXURE I (SOURCE CODE)

```
import numpy as np

import pandas as pd

import tensorflow as tf

from tensorflow.keras.models import Model

from tensorflow.keras.layers import Input, Dense, Bidirectional, LSTM, Concatenate, Flatten

from sklearn.model_selection import train_test_split

from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error

import matplotlib.pyplot as plt

from bayes_opt import BayesianOptimization

import seaborn as sns

from scipy.stats import norm

import shap

import lime.lime_tabular




# Load Data

def load_data():

    train_data = pd.read_csv("Training.csv")

    test_data = pd.read_csv("Testing.csv")

    for col in ["Close", "Open", "High", "Low"]:
```

```
        train_data[col] = pd.to_numeric(train_data[col], errors="coerce")
        test_data[col] = pd.to_numeric(test_data[col], errors="coerce")
    train_data.dropna(inplace=True)
    test_data.dropna(inplace=True)
    return train_data, test_data



train_data, test_data = load_data()



# Prepare input and output
X_train_full = train_data[["Open", "High", "Low"]].values
y_train_full = train_data["Close"].values
X_test = test_data[["Open", "High", "Low"]].values
y_test = test_data["Close"].values



X_train, X_val, y_train, y_val = train_test_split(X_train_full,
y_train_full, test_size=0.2, random_state=42)



# Reshape input for Bi-LSTM
X_train = X_train.reshape((X_train.shape[0], 1, X_train.shape[1]))
```

```
X_val = X_val.reshape((X_val.shape[0], 1, X_val.shape[1]))

X_test = X_test.reshape((X_test.shape[0], 1, X_test.shape[1]))




# TimesNet Block

def timesnet_block(input_tensor, units=32):

    x = Dense(units, activation='relu')(input_tensor)

    x = Dense(units, activation='relu')(x)

    return x




# Bayesian Optimization Objective Function

def train_evaluate(lstm_units, timesnet_units, learning_rate, batch_size):

    lstm_units = int(lstm_units)

    timesnet_units = int(timesnet_units)

    batch_size = int(batch_size)



    # Define model

    input_tensor = Input(shape=(X_train.shape[1], X_train.shape[2]))

                          bi_lstm      =      Bidirectional(LSTM(lstm_units,
activation="relu"))(input_tensor)

    timesnet = timesnet_block(Flatten()(input_tensor), units=timesnet_units)

    merged = Concatenate()([bi_lstm, timesnet])
```

```
    output = Dense(1)(merged)


    model = Model(inputs=input_tensor, outputs=output)

model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=learning_rate
), loss="mse")


    # Train model

     model.fit(X_train, y_train, validation_data=(X_val, y_val), epochs=20,
batch_size=batch_size, verbose=0)


    # Evaluate model

    pred_val = model.predict(X_val)

    mse_val = mean_squared_error(y_val, pred_val)

    return -mse_val  # Minimize MSE




# Define Bayesian Optimizer

pbounds = {

    "lstm_units": (32, 128),

    "timesnet_units": (32, 128),

    "learning_rate": (0.0001, 0.01),

    "batch_size": (16, 64)

}
```

```
optimizer = BayesianOptimization(f=train_evaluate, pbounds=pbounds,
random_state=42)

optimizer.maximize(init_points=5, n_iter=10)




# Best Parameters

best_params = optimizer.max["params"]

best_params["lstm_units"] = int(best_params["lstm_units"])

best_params["timesnet_units"] = int(best_params["timesnet_units"])

best_params["batch_size"] = int(best_params["batch_size"])




print("Best Parameters:", best_params)




# Train Final Model with Best Params

input_tensor = Input(shape=(X_train.shape[1], X_train.shape[2]))

bi_lstm = Bidirectional(LSTM(best_params["lstm_units"],
activation="relu"))(input_tensor)

timesnet = timesnet_block(Flatten()(input_tensor),
units=best_params["timesnet_units"])

merged = Concatenate()([bi_lstm, timesnet])

output = Dense(1)(merged)
```

```
final_model = Model(inputs=input_tensor, outputs=output)

final_model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=best_pa
rams["learning_rate"]), loss="mse")

final_model.fit(X_train, y_train, validation_data=(X_val, y_val), epochs=40,
batch_size=best_params["batch_size"], verbose=1)




# Make Predictions

pred_test = final_model.predict(X_test)




# Evaluate

mae_test = mean_absolute_error(y_test, pred_test)

mse_test = mean_squared_error(y_test, pred_test)

rmse_test = np.sqrt(mse_test)

mape_test = mean_absolute_percentage_error(y_test, pred_test)

print(f"Test MAE: {mae_test}, MSE: {mse_test}, RMSE: {rmse_test}, MAPE:
{mape_test * 100}%")




# Plot Results

plt.figure(figsize=(10, 5))

plt.plot(y_test, label="Actual", color="blue")
```

```
plt.plot(pred_test, label="Predicted", color="red")

plt.legend()

plt.title("Actual vs Predicted (Test)")

plt.xlabel("Time")

plt.ylabel("Closing Price")

plt.show()




# Compute prediction errors

errors = (y_test - pred_test).flatten()  # Ensure it's 1D




# Fit a normal distribution to the errors

mu, sigma = np.mean(errors), np.std(errors)




# Create the figure

plt.figure(figsize=(8, 5))




# Plot histogram with clear bin separation

plt.hist(errors,  bins=30,  color="purple",  alpha=0.6,  edgecolor="black",
linewidth=1.2, density=True)
```

```
# Generate x values for the bell curve
x = np.linspace(min(errors), max(errors), 1000)
y = norm.pdf(x, mu, sigma)  # Gaussian curve


# Plot the bell curve
plt.plot(x, y, color="blue", linewidth=2)
# Labels and title
plt.xlabel("Prediction Error", fontsize=12)
plt.ylabel("Density", fontsize=12)
plt.title("Error Distribution Curve", fontsize=14)


# Grid for better readability
plt.grid(axis="y", linestyle="--", alpha=0.6)


# Show legend
plt.legend()
```

```
# Show the plot

plt.show()

# Define a prediction function that reshapes input correctly

def lstm_predict(X):

      X = X.reshape(X.shape[0], 1, X.shape[1])   # Reshape to (samples,
timesteps, features)

    return final_model.predict(X, verbose=0)  # Silent prediction




# Flatten the data (SHAP KernelExplainer expects 2D input)

X_background = X_train[:50].reshape(50, 3)  # (samples, features)

X_test_flattened = X_test[:50].reshape(50, 3)




# Create SHAP KernelExplainer

explainer = shap.KernelExplainer(lstm_predict, X_background)


# Compute SHAP values

shap_values = explainer.shap_values(X_test_flattened)


# Convert SHAP values to the correct format

shap_values = np.array(shap_values).squeeze()
```

```
# Summary plot
shap.summary_plot(shap_values, X_test_flattened, feature_names=['Open', 'High', 'Low'])

# Dependence plot
shap.dependence_plot(0, shap_values, X_test_flattened, feature_names=['Open', 'High', 'Low'])

# Force plot for a single prediction
shap.force_plot(explainer.expected_value, shap_values[0], X_test_flattened[0], feature_names=['Open', 'High', 'Low'])

# Custom wrapper for LIME compatibility
def lime_prediction_wrapper(X):
    # Reshape back to 3D shape expected by LSTM
    X_reshaped = X.reshape(X.shape[0], 1, X.shape[1])
    return final_model.predict(X_reshaped).flatten()

# Create a LIME explainer
def create_lime_explainer(X_train):
    explainer = lime.lime_tabular.LimeTabularExplainer(
        training_data=X_train.reshape(X_train.shape[0], -1),  # Flatten for tabular format
```

```
        mode='regression',
        feature_names=['Open', 'High', 'Low'],
        class_names=['Close'],
        discretize_continuous=False
    )
    return explainer

# Create explainer
lime_explainer = create_lime_explainer(X_train)

# Select a sample instance to explain
sample_index = 10
sample_instance = X_test[sample_index].reshape(1, -1)

# Prediction for the chosen sample
sample_prediction = final_model.predict(sample_instance.reshape(1, 1, -1))[0][0]
print(f"Actual Close Price: {y_test[sample_index]}")
print(f"Predicted Close Price: {sample_prediction}")

# Generate explanation
explanation = lime_explainer.explain_instance(
    sample_instance.flatten(),  # Flatten for LIME compatibility
```

```
    lime_prediction_wrapper  # Custom wrapper for LSTM model
)


# Visualize the explanation
explanation.show_in_notebook()
```